

APPENDIX D

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Emre Alptekin et al.	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	FIS920150066US1	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	8278200	B2	2012-10-02	Lavoie et al.	

If you wish to add additional U.S. Patent citation information please click the Add button.

Add

U.S.PATENT APPLICATION PUBLICATIONS						Remove
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1					

If you wish to add additional U.S. Published Application citation information please click the Add button.

Add

FOREIGN PATENT DOCUMENTS								Remove
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐

If you wish to add additional Foreign Patent Document citation information please click the Add button

Add

NON-PATENT LITERATURE DOCUMENTS				Remove
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Emre Alptekin et al.	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	FIS920150066US1	

	1	Filed Application titled "Metal Semiconductor Alloy Contact Resistance Improvement," US Application No. 14/172,525 filed February 4, 2014, Attorney Docket No. FIS920130170US1, 37 pages.	☐
--	---	--	--------------------------

If you wish to add additional non-patent literature document citation information please click the Add button **Add**

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	Emre Alptekin et al.
Art Unit	
Examiner Name	
Attorney Docket Number	FIS920150066US1

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Amanda J. Smith/	Date (YYYY-MM-DD)	2015-06-03
Name/Print	Amanda J. Smith	Registration Number	71,112

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

First Named Inventor/Applicant Name:

Emre Alptekin

Filer:

Amanda J Smith

Attorney Docket Number:

FIS920150066US1

Filed as Large Entity

Filing Fees for Utility under 35 USC 111(a)

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	280	280
Utility Search Fee	1111	1	600	600
Utility Examination Fee	1311	1	720	720

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1600

Electronic Acknowledgement Receipt

EFS ID:	22521615
Application Number:	14729298
International Application Number:	
Confirmation Number:	2837
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	45094
Filer:	Amanda J Smith
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US1
Receipt Date:	03-JUN-2015
Filing Date:	
Time Stamp:	12:44:32
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 1600
RAM confirmation Number	10494
Deposit Account	090458
Authorized User	WARNICK, SPENCER K.

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.20 (Post Issuance fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		FIS920150066US1_Application.pdf	138061	yes	23
			54d97f1fd89a64025c530b26ea3afe2da795d7c8		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	16	
	Claims		17	22	
	Abstract		23	23	
Warnings:					
Information:					
2	Drawings-only black and white line drawings	FIS920150066US1_FormaDrawings.pdf	69799	no	5
			e26939b78fccfb6f8de27b670b32f34e9777b45c		
Warnings:					
Information:					
3	Application Data Sheet	FIS920150066US1_Application DataSheet.pdf	1561566	no	8
			afa5c41cf1c50d6d5058108d0bbc6f8a54dea422		
Warnings:					
Information:					
4	Oath or Declaration filed	FIS920150066US1_Filed_Dec-Assign.pdf	789132	no	8
			7d752b991ca402f5fe8a9d6f368d3586d0f177cb		
Warnings:					
Information:					
5	Power of Attorney	FIS920150066US1_Transmittal-SignedPOA.pdf	321840	no	2
			cde269c915f67a3b0f8b358cd922c2c8cbd66471		
Warnings:					
Information:					
6	Non Patent Literature	FIS920130170US1_FiledApplication.pdf	202378	no	37
			1625017f36ac2a62c649570cbc7acd1d22fb2652		

Warnings:					
Information:					
7	Information Disclosure Statement (IDS) Form (SB08)	FIS920150066US1_IDSPTOSB08a.pdf	612289 f88f520c25f0a236dac084a348a842732885b33f	no	4
Warnings:					
Information:					
8	Fee Worksheet (SB06)	fee-info.pdf	35050 33947694364003f7e134789572e747d9774163d2	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			3730115		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

BACKGROUND

Technical Field

[0001] The present invention relates to contacts over a semiconductor substrate and methods of forming the same. More specifically, the present invention relates to forming an intermix region at an interface of a bottom portion of contacts and an upper portion of the semiconductor substrate, respectively.

Related Art

[0002] State of the art field effect transistors (FETs) can be fabricated by depositing a gate conductor over a gate dielectric and a semiconductor substrate. Generally, the FET fabrication process implements lithography and etching processes to define the gate structures. After providing the gate structures, source/drain extensions may be formed into a portion of the semiconductor substrate and on both sides of each gate structure by ion implantation.

Sometimes this implant is performed using a spacer to create a specific distance between the gate structure and the implanted junction. Source and drain may be doped with a p-type or n-type dopant. The term "p-type" refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. "N-type" refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. After formation of the transistor, a set of contact openings may be created to expose the source and drain. The contact openings may be coated with a Ti-based silicide liner. Subsequently, contact openings may be filled with a metal

material, such as W.

[0003] In a p-type FET (PFET), performance with a Ti-based silicide liner is a challenge for 14/10 nm silicon-on-insulator (14/10 SOI) technologies. In order to improve workfunction of Ti-based silicide in the source and drain of a PFET, a bilayer of NiPt and Ti has been employed to coat contact openings over the source and drain. However, this bilayer requires deposition of a relatively thick metal layer. As semiconductor devices get smaller, requiring a thick bilayer in the contact causes contact opening fill issues, such as an incomplete contact fill.

SUMMARY

[0004] A first aspect of the invention includes a method for forming a contact in a dielectric layer to a semiconductor substrate. The method comprises: forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

[0005] A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain. The method comprises: forming a contact opening in the dielectric layer to expose an upper portion of each of the source and the drain; depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of

each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain; depositing a refractory metal layer over the first intermix region; and depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.

[0006] A third aspect of the invention includes a semiconductor structure including a contact to a semiconductor substrate. The contact comprises: a contact opening in a dielectric layer over the semiconductor substrate; a first liner layer substantially coating a sidewall of the contact opening; a refractory metal layer substantially coating the first liner layer in the contact opening; a metal substantially filling the contact opening; and an intermix region at an upper portion of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The embodiments of this invention will be described in detail, with reference to the following figures, wherein like designations denote like elements, and wherein:

[0008] FIGS. 1-8 show cross-sectional views of a semiconductor structure undergoing processing according to embodiments of the invention.

[0009] FIG. 9 shows a cross-sectional view of another embodiment of the semiconductor structure according to embodiments of the invention.

DETAILED DESCRIPTION

[0010] Aspects of the present invention utilize an intermix concept as a template for formation of a contact to a semiconductor substrate of a semiconductor structure. In some embodiments, the semiconductor structure may be a transistor, e.g., PFET or n-type FET (NFET) having a gate, a source, and a drain as is known in the art. In other embodiments, semiconductor structure may be a wire, a FinFet, or a hetero-bipolar junction transistor (HBT) or other silicon based microelectronic devices. One aspect of the present invention involves depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate thereunder, e.g., source and drain. Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved.

[0011] Referring now to FIG. 1, a semiconductor structure 100 is shown including a semiconductor substrate 110, an isolation region 112, and a gate structure 114 within a dielectric layer 120 over substrate 112. Substrate 112 may be employed as a semiconductor layer of a semiconductor-on-insulator substrate (not shown), or a bulk substrate (not shown) as understood in the art. Substrate 112 may include any now known or later developed semiconductor material including but not limited to silicon, germanium, silicon germanium, silicon carbide, and those consisting essentially of one or more III-V compound semiconductors having a composition defined by the formula $Al_{X1}Ga_{X2}In_{X3}As_{Y1}P_{Y2}N_{Y3}Sb_{Y4}$, where $X1$, $X2$, $X3$, $Y1$, $Y2$, $Y3$, and $Y4$ represent relative proportions, each greater than or equal to zero and

$X1+X2+X3+Y1+Y2+Y3+Y4=1$ (1 being the total relative mole quantity). Other suitable substrates include II-VI compound semiconductors having a composition $Zn_{A1}Cd_{A2}Se_{B1}Te_{B2}$, where A1, A2, B1, and B2 are relative proportions each greater than or equal to zero and $A1+A2+B1+B2=1$ (1 being a total mole quantity). Furthermore, a portion or entire substrate may be strained. Substrate 112 may be doped, undoped or contain doped and undoped regions therein.

[0012] In some embodiments of the invention, substrate 110 may include at least one isolation region 112 therein. Isolation region 112 may be a trench isolation region or a field oxide isolation region as is known in the art. A trench isolation region can be formed utilizing a conventional trench isolation process well known to those skilled in the art. For example, lithography, etching and filling of the trench with a trench dielectric such as an oxide may be used in forming the trench isolation region. Optionally, a liner may be formed in the trench prior to trench fill. After the trench is filled, a densification step may be performed as well as a planarization process. A field oxide isolation region may be formed utilizing a so-called local oxidation of silicon process. The at least one isolation region 112 may provide isolation between neighboring gate structures, typically required when the neighboring gate structures have opposite conductivities, i.e., n-type transistors and p-type transistors. As such, the at least one isolation region 112 can separate, for example, a first device region in which an n-type transistor device can be formed and second device region in which a p-type transistor device can be formed.

[0013] Further, FIG. 1 illustrates that a gate 114 may be formed on substrate 112. Gate structure 114 may include a gate dielectric material 116 and a gate conductor 118. Gate structure 114 may be used to control output current of the semiconducting device through electrical or magnetic fields. Gate dielectric 116 may include, but is not limited to: HfO_2 , ZrO_2 , La_2O_3 , Al_2O_3 ,

TiO₂, SrTiO₃, LaAlO₃, Y₂O₃, HfO_xN_y, ZrO_xN_y, La₂O_xN_y, Al₂O_xN_y, TiO_xN_y, SrTiO_xN_y, LaAlO_xN_y, Y₂O_xN_y, SiON, SiN_x, a silicate thereof, and an alloy thereof. Each value of x is independently from 0.5 to 3 and each value of y is independently from 0 to 2. In some embodiments, a multilayered gate dielectric structure comprising different gate dielectric materials, e.g., silicon dioxide, and a dielectric metal oxide can be formed and used as gate dielectric material 116. Gate conductor 118 may include, but is not limited to: doped polysilicon, doped silicon germanium, an elemental metal (e.g., tungsten, titanium, tantalum, aluminum, nickel, ruthenium, palladium and platinum), an alloy of at least two elemental metals, an elemental metal nitride (e.g., tungsten nitride, aluminum nitride, and titanium nitride), or multilayered combinations thereof.

[0014] Gate structure 114 may be formed by depositing gate dielectric 116 on substrate 112. As used herein, “depositing” may include any now known or later developed techniques appropriate for the material to be deposited including but are not limited to, for example: chemical vapor deposition (CVD), low-pressure CVD (LPCVD), plasma-enhanced CVD (PECVD), semi-atmosphere CVD (SACVD) and high density plasma CVD (HDPCVD), rapid thermal CVD (RTCVD), ultra-high vacuum CVD (UHVCVD), limited reaction processing CVD (LRPCVD), metalorganic CVD (MOCVD), sputtering deposition, ion beam deposition, electron beam deposition, laser assisted deposition, thermal oxidation, thermal nitridation, spin-on methods, physical vapor deposition (PVD), atomic layer deposition (ALD), chemical oxidation, molecular beam epitaxy (MBE), plating, evaporation. Additionally, subsequent to the deposition of gate dielectric 116, gate conductor 118 can be deposited on gate dielectric 116. The precise thickness of gate dielectric 116 and gate conductor 118 may vary widely with the intended application.

[0015] Gate dielectric 116 and gate conductor 118 can be patterned into a functional gate structure 114. Gate structure 114 can be formed by lithography and etching. Lithography can include forming a photoresist (not shown) on the topmost surface of gate conductor 118 exposing the photoresist to a desired pattern of radiation, and then developing the exposed photoresist with a resist developer to provide a patterned photoresist atop the gate conductor 118. At least one etch is then employed which transfers the pattern from the patterned photoresist into the various materials. As used herein, "etching" may include any now known or later developed techniques appropriate for the material to be etched including but not limited to, for example: isotropic etching, anisotropic etching, plasma etching, sputter etching, ion beam etching, reactive-ion beam etching and reactive-ion etching (RIE).

[0016] Still referring to FIG. 1, a spacer 122 may be formed on each sidewall of gate 114. Spacers 122 may be formed by depositing a spacer material and etching the spacer material. Spacers 122 may include a spacer dielectric material as is known in the art, such as, but not limited to: a dielectric oxide, dielectric nitride, and/or dielectric oxynitride. Additionally, a source 124a and drain 124b may be formed in a portion of substrate 110 on either side of gate 114. In some embodiments, a source extension region and a drain extension region (not separately shown) may be formed prior to forming spacer 122 utilizing an extension ion implantation process. As such, a portion of each of the source extension region and the drain extension region would be located beneath the spacer 122. After forming the spacers 122, source 124a and drain 124b are formed utilizing a source/drain ion implantation process. An activation anneal may follow the implantation processes. Source 124a (including the corresponding source extension region) and drain 124b (including the corresponding drain extension region) may be doped with a p-type or n-type dopant. The term "p-type" refers to the addition of impurities to an

intrinsic semiconductor that creates deficiencies of valence electrons. In a silicon-containing substrate, examples of p-type dopants, i.e., impurities, include, but are not limited to, boron, aluminum, gallium and indium. "N-type" refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. In a silicon-containing substrate, examples of n-type dopants, i.e., impurities, include, but are not limited to, antimony, arsenic and phosphorous. The concentration of dopants within source 124a (and the corresponding source extension region) and drain 124b (and the corresponding drain extension region) can be within ranges known in the art.

[0017] Still referring to FIG. 1, a dielectric layer 120 may be formed such that dielectric layer 120 substantially surrounds gate 114. In one embodiment of the invention, formation of dielectric layer 120 may include deposition of a dielectric material 126 and a dielectric cap material 128. While shown as two separate layers, it is to be understood that dielectric layer 120 is not limited to only two layers. Rather, in other embodiments of the invention, dielectric layer 120 may include a single layer or more than two separate layers. Dielectric material 126 may include, but is not limited to: silicon nitride (Si_3N_4), silicon oxide (SiO_2), fluorinated SiO_2 (FSG), hydrogenated silicon oxycarbide (SiCOH), porous SiCOH , boro-phospho-silicate glass (BPSG), silsesquioxanes, carbon (C) doped oxides (i.e., organosilicates) that include atoms of silicon (Si), carbon (C), oxygen (O), and/or hydrogen (H), thermosetting polyarylene ethers, SiLK (a polyarylene ether available from Dow Chemical Corporation), a spin-on silicon-carbon containing polymer material available from JSR Corporation, other low dielectric constant (<3.9) material, or layers thereof. The upper surface of gate 114 is exposed immediately after depositing dielectric material 126. In some embodiments, a planarization process or an etch back process follows the deposition of the dielectric material 126. As is illustrated, dielectric material 126 may have an upper surface that is coplanar with an upper surface of gate 114. Dielectric

material 126 may have a bottom surface that contacts an upper surface of source 124a, an upper surface of the drain 124b, and an upper surface of insulation regions 112. The thickness of the dielectric material 126 that can be employed in the present application may vary depending on the type of dielectric material employed as well as the method that was employed in forming the same. In one embodiment, dielectric material 126 has a thickness from 80 nm to 500 nm. Other thicknesses that are greater or lesser than the range provided above can also be used for dielectric material 126.

[0018] Subsequently, dielectric cap material 128 may be deposited on dielectric material 126. Dielectric cap material 128 may include an insulator material that differs from dielectric material 126. Examples of insulator materials that can be used as dielectric cap material 128 include, for example, silicon dioxide, silicon nitride and silicon oxynitride. Dielectric cap material 128 may have a thickness from 1 nm to 20 nm. Other thicknesses can also be used for the thickness of the dielectric cap material 128.

[0019] As shown in FIG. 2, a plurality of contact openings 130a, 130b may be formed in a dielectric layer 120 to substrate 110. That is, contact openings 130a, 130b may be formed through dielectric cap material 128 and dielectric material 126 to expose source 124a and drain 124b. For example, contact opening 130a exposes source 124a, while contact opening 130b exposes drain 124b. Contact openings 130a, 130b can be formed by lithography and etching. Portions of dielectric material 126 and dielectric cap material 128 remain on the upper surfaces of gate 114. As is shown in the drawings of the present application, a portion of dielectric cap material 128 that is located on the upper surface gate structure 114 may extend beyond the edges of gate structure 114 and beyond an outermost edge of each spacer 122. Further, dielectric cap material 128 and dielectric material 126 that are present above isolation regions 112 extend

beyond the outermost edges of isolation regions 112. In some embodiments, plurality of openings 130a, 130b have a width (measured from one exposed sidewall of one portion of dielectric material 126 to another exposed sidewall of another portion of dielectric material 126) of from 10 nm-50 nm. Other widths that are greater or lesser than the aforementioned widths can also be used in the present application.

[0020] As shown in FIG. 3, a first liner layer 132 may be deposited to conformally coat each contact opening 130a, 130b and substantially cover dielectric cap material 128. First liner layer 132 may be deposited using ionized PVD or radio frequency PVD (RFPVD) or other directional energetic thin film deposition techniques such as Ion beam assisted deposition, performed at room temperature or above. First liner layer 132 may include, but is not limited to, at least one of: NiPt, Ni, Ti, TiSi, Co, Er, Pt, and Yb. First liner layer 132 may be deposited such that a first intermix region 134a, 134b is formed at an interface of the bottom portion of each contact opening 130a, 130b and the upper portion of each of source 124a and drain 124b.

[0021] First intermix regions 134a, 134b may originate from the high energy of the incoming sputtered atoms (from first liner layer 132) which are thus pushed into the Si lattice of source 124a and drain 124b. Therefore, first intermix regions 134a, 134b may include a mixture of the material used for first liner layer 132 and a material used for the respective semiconductor substrate 110 thereunder. In the embodiment shown in FIG. 3, first intermix regions 134a, 134b may have a composition including a material used for first liner layer 132 and materials used for the respective source and drain 124a, 124b thereunder. While the Si lattice is a dense way to pack Si atoms, it has a relatively open structure due to its bonding being covalent. The first few metal atoms, arriving with relatively high energy, can be forced into the interstitial sites of the Si lattice to create a thin region (intermix regions 134a, 134b) of mixed Si and metal on the order of

3-5 nm in thickness, for example. In some cases, first intermix region 134a, 134b may have a relatively high Si content (greater than 50% Si in atomic composition). In other embodiments, the compositions of first intermix regions 134a, 134b may vary depending on one or more factors including, but not limited to: the type of material being deposited, the composition of source 124a and drain 124b, the energy of the atoms of the material being deposited, and/or the conditions of the deposition technique (temperature, pressure, reaction, atmosphere, etc.). In some embodiments, it may be desirable to select a target composition of first intermix regions 134a, 134b in view of one or more criteria, e.g., etch selectivity. First intermix region 134a, 134b may be amorphous and a subsequent anneal may crystalize intermix region 134a, 134b into a lower resistance layer as will be described herein.

[0022] First intermix regions 134a, 134b allow for formation of a very thin contact layer that is normally very difficult to control with standard sputtering tools. First intermix regions 134a, 134b are much more conformal on Si structures that are not planar. Additionally, upon annealing of first intermix regions 134a, 134b (which will be described further herein), very little if any metal diffusion is needed to form the low resistance contact layer which is often a source of defects in small devices. Encroachment of silicide under spacers 122 and encroachment on defective silicon areas may be substantially reduced or eliminated by this intermixing concept, thus increasing the yield of semiconductor structures or devices.

[0023] Referring now to FIG. 4, another step of the method according to embodiments of the invention is shown. In this step, first liner layer 132 is removed from each contact opening 130a, 130b as well as from the upper surface of dielectric cap material 128. That is, the remaining portion of first liner layer 132 which does not diffuse into the respective source 124a and drain 124b is removed. First liner layer 132 may be removed by etching using one of the various

etching techniques previously described. In some embodiments, first liner layer 132 may be removed by an aqua regia etch process. Aqua regia etching is a wet etch process and involves providing a solution of hydrochloric and nitric acids to the surfaces to be etched. First liner layer 132 may be removed such that intermix regions 134a, 134b are exposed at the bottom portion of contact openings 130a, 130b.

[0024] As shown in FIG. 5, another aspect of the method according to embodiments of the invention is shown to include depositing a refractory metal layer 140 to conformally coat each contact opening 130a, 130b and substantially cover dielectric cap material 128 and first intermix regions 134a, 134b at the bottom portion of each contact opening 130a, 130b. Refractory metal layer 140 may include, but is not limited to: W, Ti, Ta or Mo employed directly as a “metal”; a refractory metal employed as a constituent of a “compound” thereof, e.g., TiN, TaN, etc.; or even as a constituent of an “alloy” with another metal, e.g., titanium-tungsten. In some embodiments, as shown in FIG. 5, refractory metal layer 140 may include a bilayer of a first refractory material 142 and a second refractory material 144 such that first refractory metal material 142 substantially separates first liner layer 132 from second refractory metal layer 144. First and second refractory metal materials 142, 144 may include any of the refractory metal materials previously mentioned. For example, in some embodiments, first refractory metal material 142 may include Ti and second refractory metal material 144 may include TiN. In this example, first refractory metal material 142 may be deposited using RFPVD and second refractory metal material 144 may be deposited using ALD. While shown as a bilayer, it is to be understood that refractory metal layer 140 may include any number of layers of refractory metal materials without departing from aspects of the invention.

[0025] Referring now to FIG. 6, a metal 150 may be deposited over dielectric cap material 128

and within each contact opening 130a, 130b to substantially coat refractory metal layer 140 therein and substantially fill contact openings 130a, 130b. Alternatively, a plating process that fills each contact opening 130a, 130b from the bottom upwards can be used. In an embodiment of the invention where refractory metal layer 140 includes a bilayer, second refractory metal material 144 may substantially separate first refractory metal material 142 from metal layer 150. Metal layer 150 may include, but is not limited to: W, Co, Cu, and Al. As shown in FIG. 7, semiconductor structure 100 may undergo a planarization process such as, for example, chemical mechanical polishing (CMP) to create contacts 170a, 170b within contact openings 130a, 130b. During the planarization process, portions of refractory metal layer 140 and metal layer 150 that extend beyond the upper surface of dielectric cap material 128 are removed such that refractory metal layer 140 and metal layer 150 within contact openings 130a, 130b are coplanar with the upper surface of dielectric cap material 128.

[0026] In one example, contacts 170a and 170b may include NiPt as first liner layer 132, a Ti/TiN bilayer as refractory metal layer 140, and W as metal 150. In this embodiment, first intermix regions 134a, 134b may include a NiPt silicide and second intermix regions 154a, 154b may include a NiPtTi silicide. However, it is to be understood that other combinations of materials may be used to create contacts 170a, 170b without departing from aspects of the invention. For example, in another embodiment, first liner layer 132 may include Ni, refractory metal layer 140 may include a Ti/TiN bilayer, and metal 150 may include W. In this embodiment, first intermix regions 134a, 134b may include a Ni silicide and second intermix regions 154a, 154b may include a NiTi silicide.

[0027] As shown in FIG. 8, a laser anneal process may be performed such that a second intermix region 154a, 154b is formed at the interface of the bottom portion of contacts 170a,

170b and upper portion of source 124a and drain 124b. Second intermix regions 154a, 154b may have a thickness of 3-5 nm. The laser anneal process may transform the composition of first intermix region 134a, 134b (FIGS. 3-7) into a second composition resulting in second intermix region 154a, 154b in place of first intermix region 134a, 134b. That is, the laser anneal process may crystalize intermix regions 134a, 134b into a lower resistance layer, i.e. a second intermix region 154a, 154b. Second intermix regions 154a, 154b may have compositions including a mixture of the materials used for source 124a and drain 124b, first liner layer 132, and refractory metal layer 140. In an embodiment where Ti is used as refractory metal layer 140, second intermix region 154a, 154b may be a silicide, such as but not limited to: TiSi, NiSi, or NiPtTiSi. That is, during the laser anneal process, refractory metal 140 may diffuse into first intermix regions 134a, 134b (FIGS. 3-7) to create second intermix regions 154a, 154b. The laser anneal process may be performed at temperature of approximately 700°C-1100°C for between approximately 0.1 milliseconds to approximately 1 millisecond. More particularly, the laser anneal process may be performed at approximately 800°C for approximately 0.25 milliseconds. In some embodiments, the anneal temperature and duration may be dependent on the materials being used. In one embodiment, the anneal can utilize a laser having an energy from 500 watts to 2000 watts. In another embodiment, the anneal can be performed utilizing a raster scan beam size of from 1 cm long to 100 microns wide. In one example, the anneal can be performed utilizing a 10.6 micron CO₂ laser. In order to promote the efficiency of the CO₂ laser coupling with the semiconductor substrate, a laser with a shorter wavelength (e.g., InGaAs laser at 880 nm) can be used in conjunction with the CO₂ laser.

[0028] Referring now to FIG. 9, another embodiment of the invention is shown. In this embodiment, first liner layer 132 is not removed prior to depositing refractory metal layer 140.

In this embodiment, first liner layer 132 is deposited as described with respect to FIG. 3, such that the deposition of first liner layer 132 creates first intermix region 134a, 134b. However, after deposition of first liner layer 132, refractory metal layer 140 is deposited as described with respect to FIG. 5. While refractory metal layer 140 is shown as a bilayer, refractory metal layer 140 may include any number of layers of refractory metal materials, as previously discussed. Further, this embodiment also includes etching refractory metal layer 140 from the upper surface of dielectric cap layer 128 such that a portion of refractory metal layer 140 remains substantially coating the sidewalls of contact openings 130a, 130b and first intermix regions 134a, 134b at the bottom portion of contact openings 130a, 130b as described with respect to FIG. 6. Additionally, a metal layer 150 may be deposited and polished as described with respect to FIGS. 6-7. Lastly, this embodiment may also undergo the laser anneal process to create second intermix regions 154a, 154b as described with respect to FIG. 8. Second intermix regions 154a, 154b may include a mixture of the materials used for source and drain regions 124a, 124b, first liner layer 132, and refractory metal layer 140. This embodiment allows for a higher quantity of first liner layer 132 in contacts 170a, 170b due to the remaining portions of first liner layer 132 on sidewalls of contact openings 130a, 130b. Additionally, this embodiment results in a simpler process as the step of removing liner layer 132 is not performed, and therefore selectivity may not be a problem.

[0029] Contacts 170a, 170b of this embodiment may include NiPt as first liner layer 132, a Ti/TiN bilayer for refractory metal layer 140, and W as metal 150. In this embodiment, first intermix regions 134a, 134b may include NiPt silicide and second intermix regions 154a, 154b may include a NiPtTi silicide. Alternatively, first liner layer 132 may include Ti, refractory metal 140 may include TiN, and metal 150 may include W. In this embodiment, first intermix region

134a, 134b may include TiSi and second intermix region 154a, 154b may include TiSi with a higher concentration of Ti.

[0030] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0031] The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

CLAIMS

We claim:

1. A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;

depositing a first liner layer to conformally coat the contact opening;

causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;

depositing a refractory metal layer over the first intermix region; and

depositing a metal in the contact opening thereby forming the contact.

2. The method of claim 1, further comprising:

removing a remaining non-diffused portion of the first liner layer to expose the first intermix region at the upper portion of the semiconductor substrate prior to depositing the refractory metal layer.

3. The method of claim 1, wherein the depositing the first liner layer includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb).

4. The method of claim 1, wherein depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

5. The method of claim 1, wherein depositing the first liner layer includes performing physical vapor deposition.

6. The method of claim 1, further comprising:

polishing the metal; and

performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and the upper surface of semiconductor substrate, wherein the second intermix region is composed of materials used for the first intermix region and a material used for the refractory metal layer.

7. The method of claim 1, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

8. A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:

forming a contact opening in the dielectric layer to expose an upper -portion of each of the source and the drain;

depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain;

depositing a refractory metal layer over the first intermix region; and

depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.

9. The method of claim 8, further comprising:

removing a remaining non-diffused portion of the first liner layer to expose the intermix region prior to depositing the refractory metal layer.

10. The method of claim 8, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb)..

11. The method of claim 8, wherein the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

12. The method of claim 8, wherein the depositing the first liner layer includes performing physical vapor deposition.

13. The method of claim 8, further comprising:

polishing the metal in each of the openings; and

performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials used for the first intermix region and a material used for the refractory metal layer.

14. The method of claim 13, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

15. A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:

- a contact opening in a dielectric layer over the semiconductor substrate;
- a first liner layer substantially coating a sidewall the contact opening;
- a refractory metal layer substantially coating the first liner layer in the contact opening;
- a metal substantially filling the contact opening; and
- an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.

16. The contact of claim 15, wherein the intermix region includes a titanium-based silicide.

17. The contact of claim 15, wherein the first liner layer includes titanium (Ti) and the refractory metal layer includes titanium nitride (TiN).

18. The contact of claim 15, wherein the first liner layer includes nickel platinum (NiPt) and the refractory metal layer includes a bilayer having a first refractory metal material including titanium (Ti) and a second refractory metal material including titanium (TiN), the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.

19. The contact of claim 15, wherein the metal includes tungsten (W).

20. The contact of claim 15, wherein the intermix region has a thickness of approximately 3 nanometers to approximately 5 nanometers.

ABSTRACT

An aspect of the invention includes a method for forming a contact in a dielectric layer over a semiconductor substrate. The method may comprise: forming a contact opening in a dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

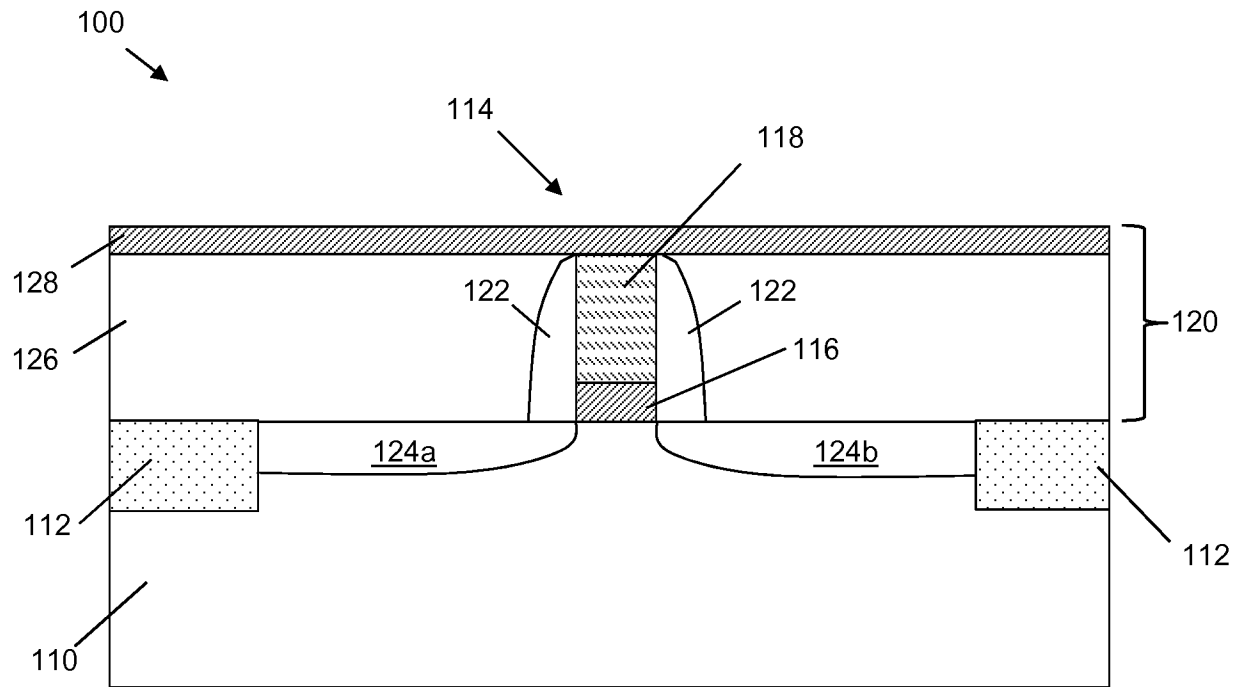


FIG. 1

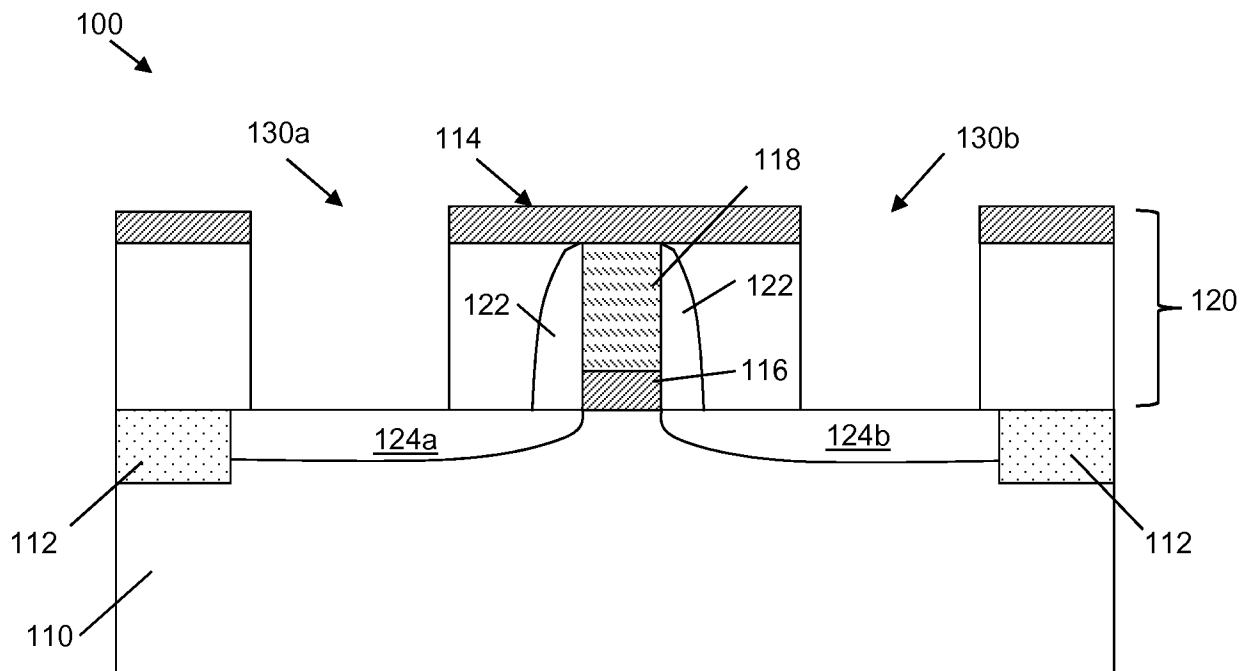


FIG. 2

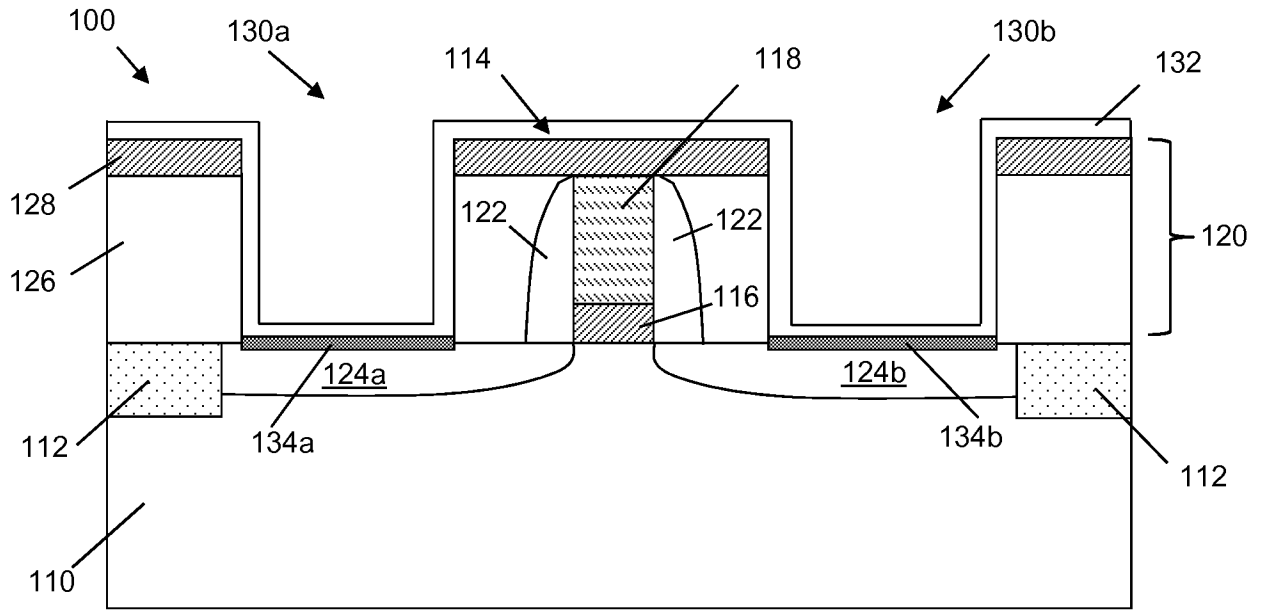


FIG. 3

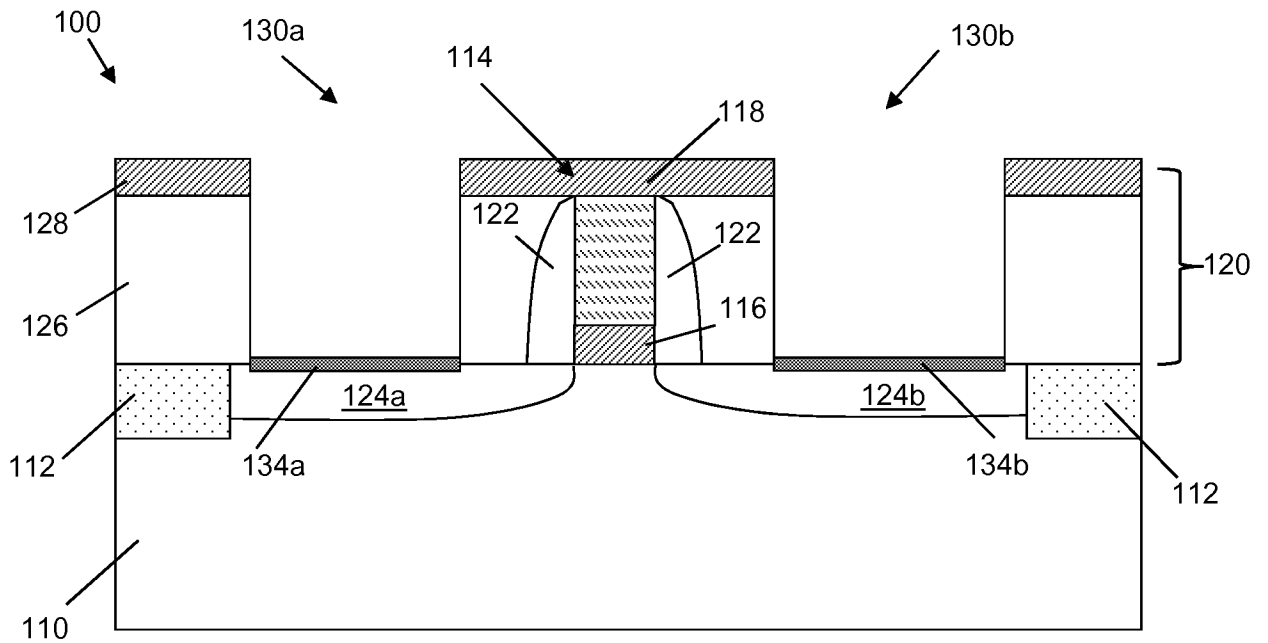


FIG. 4

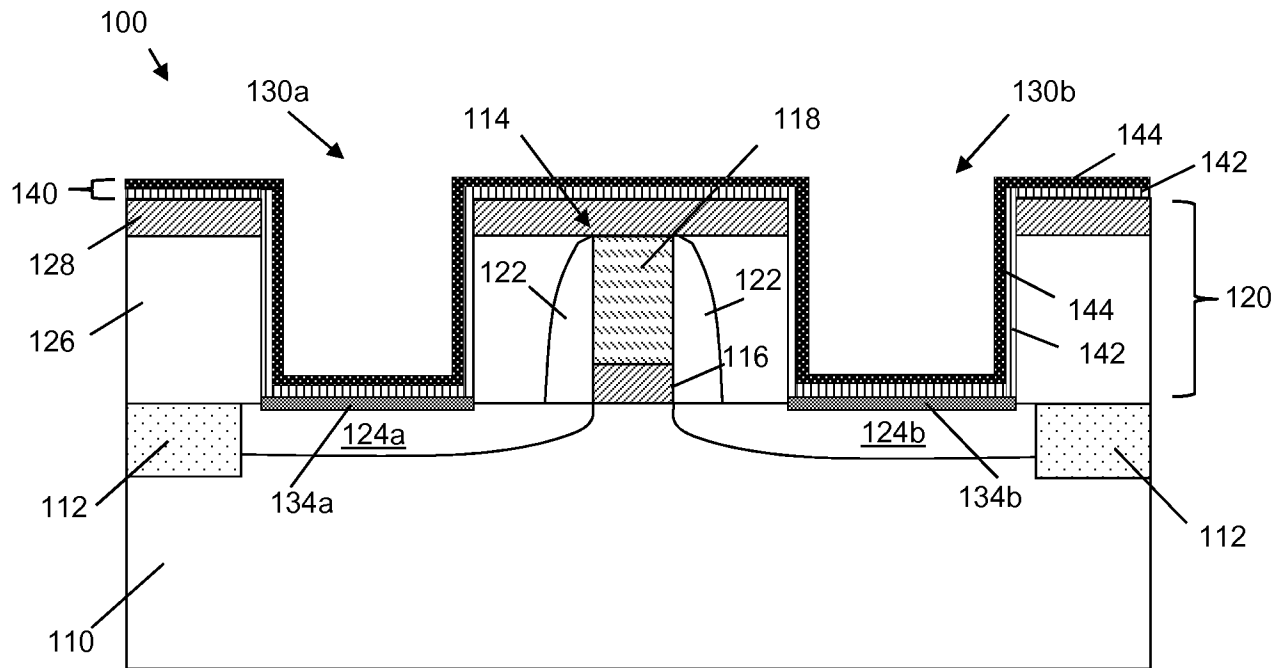


FIG. 5

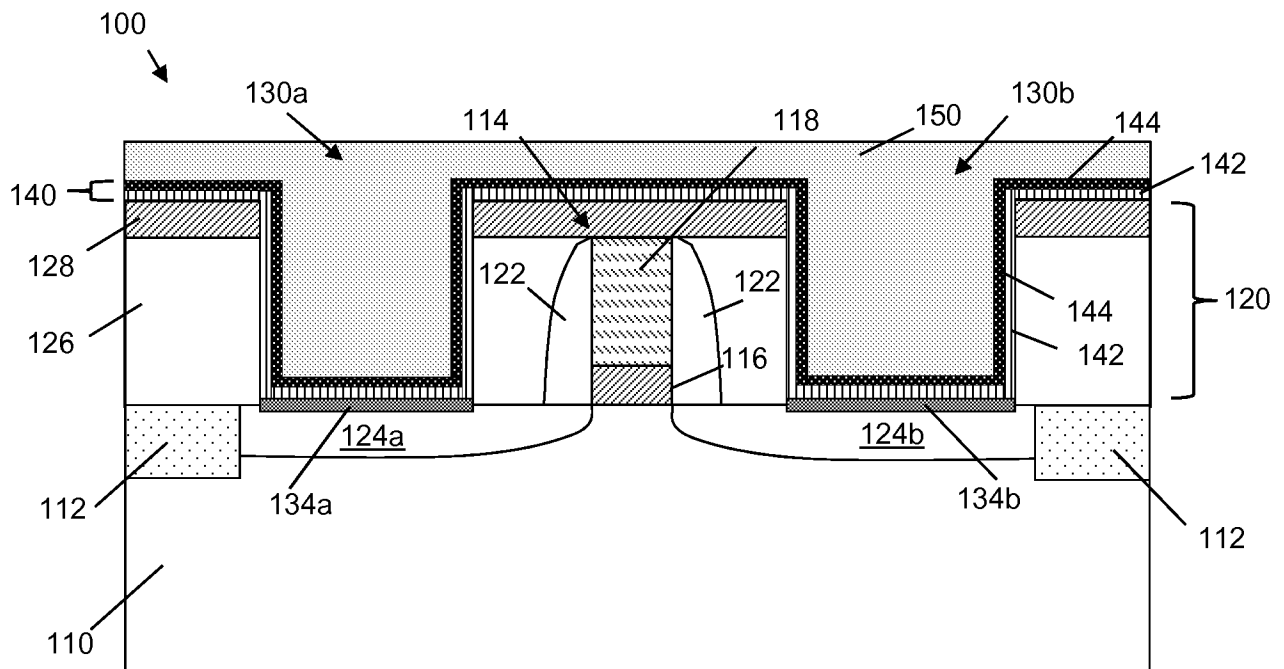


FIG. 6

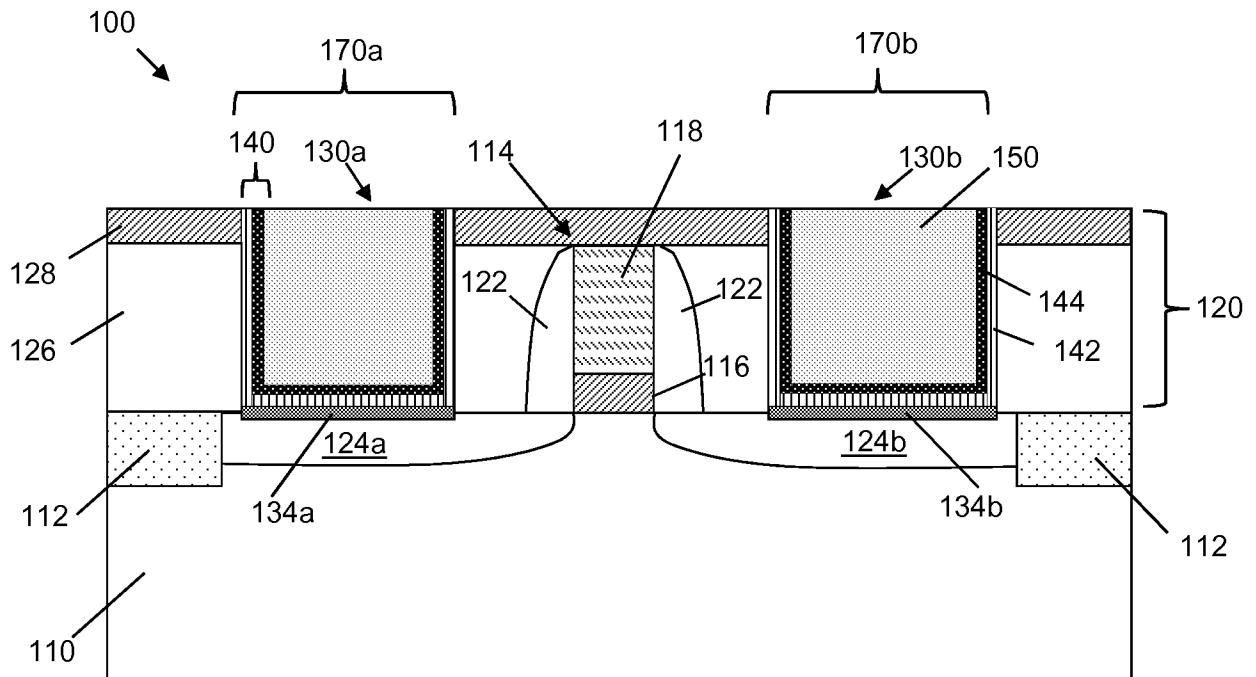


FIG. 7

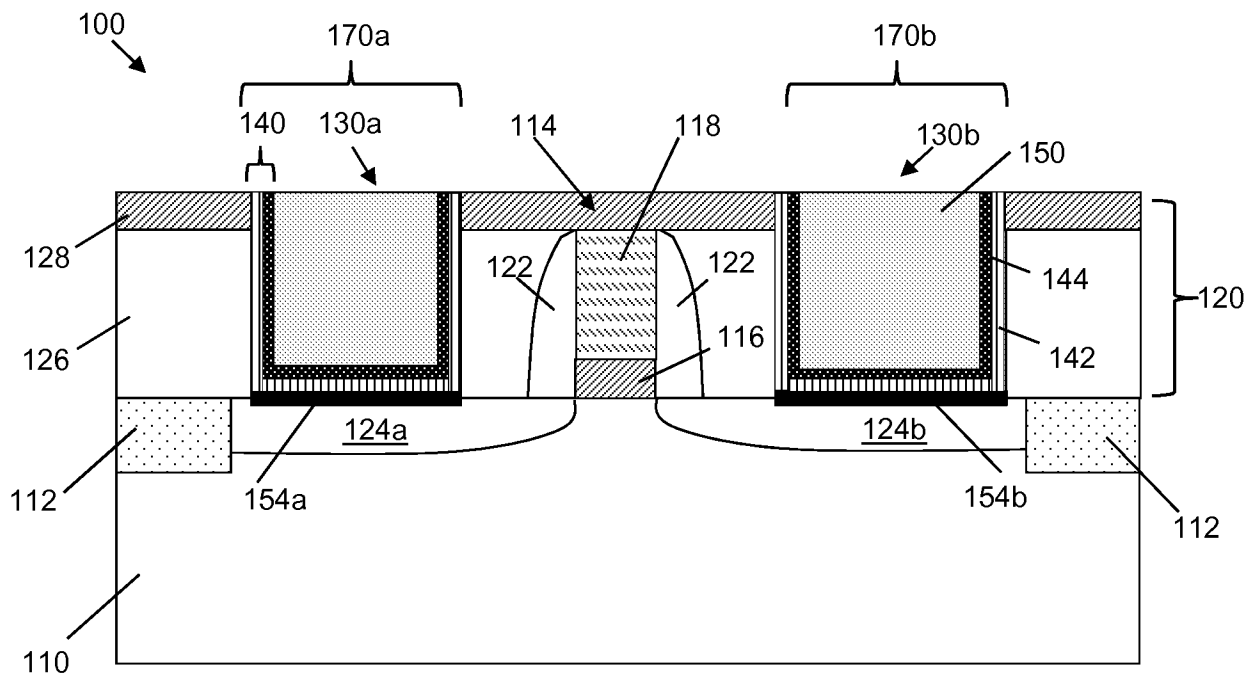


FIG. 8

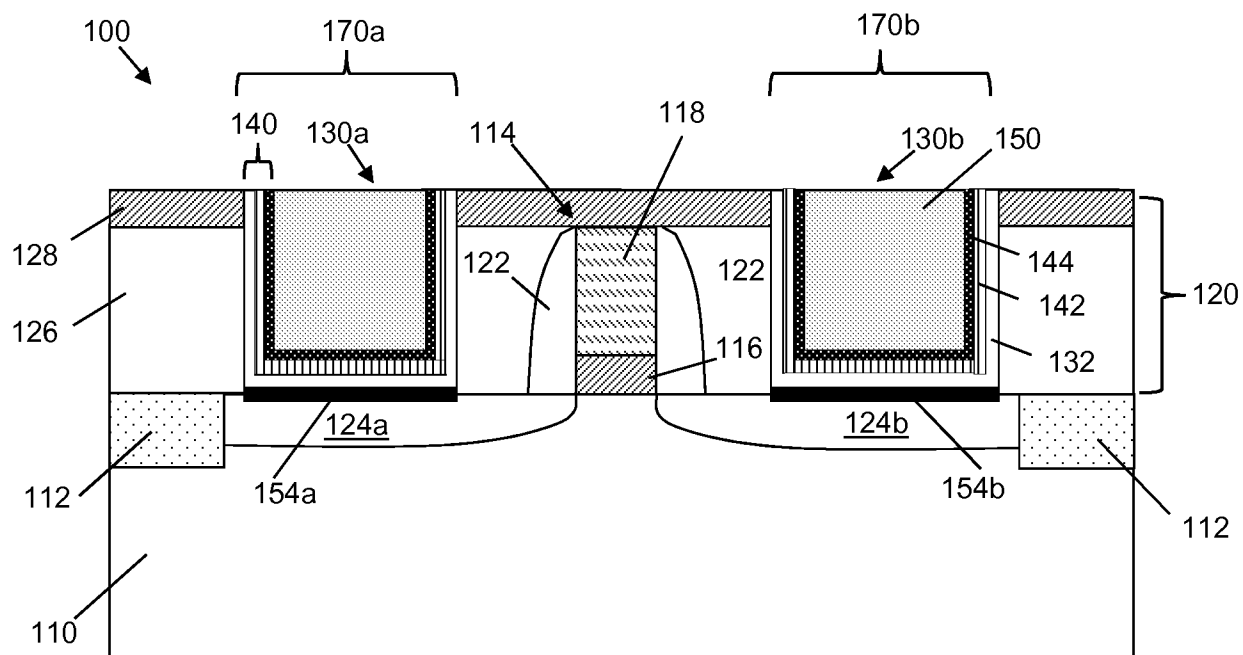


FIG. 9

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor 1					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Emre		Alptekin			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						
City	Wappingers Falls	State/Province	NY	Country of Residence i	US	
Mailing Address of Inventor:						
Address 1		2070 Route 52				
Address 2						
City	Hopewell Junction		State/Province	NY		
Postal Code	12533	Country i	US			
Inventor 2					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Nicolas	L.	Breil			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						
City	San Jose	State/Province	CA	Country of Residence i	US	
Mailing Address of Inventor:						
Address 1		3401 Iron Point Drive				
Address 2						
City	San Jose		State/Province	CA		
Postal Code	95134	Country i	US			
Inventor 3					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Christian		Lavoie			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

City	Pleasantville	State/Province	NY	Country of Residence i	US
------	---------------	----------------	----	------------------------	----

Mailing Address of Inventor:

Address 1	1101 Kitchawan Road				
Address 2	Route 134				
City	Yorktown Heights	State/Province	NY		
Postal Code	10598	Country i	US		

Inventor 4

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Ahmet	S.	Ozcan	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Chappaqua	State/Province	NY	Country of Residence i	US
------	-----------	----------------	----	------------------------	----

Mailing Address of Inventor:

Address 1	2070 Route 52				
Address 2					
City	Hopewell Junction	State/Province	NY		
Postal Code	12533	Country i	US		

Inventor 5

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Kathryn	T.	Schonenberg	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Wappingers Falls	State/Province	NY	Country of Residence i	US
------	------------------	----------------	----	------------------------	----

Mailing Address of Inventor:

Address 1	1101 Kitchawan Road				
Address 2					
City	Yorktown Heights	State/Province	NY		
Postal Code	10598	Country i	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.

Add

Correspondence Information:Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).☐ An Address is being provided for the correspondence Information of this application.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1	
		Application Number		
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME			
Customer Number	45094			
Email Address			Add Email	Remove Email

Application Information:

Title of the Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME			
Attorney Docket Number	FIS920150066US1	Small Entity Status Claimed ☐		
Application Type	Nonprovisional			
Subject Matter	Utility			
Total Number of Drawing Sheets (if any)	5	Suggested Figure for Publication (if any)		

Filing By Reference :

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	45094		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the application number blank.

Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(d). When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(h)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Remove			
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ^j (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Authorization to Permit Access:

☒ Authorization to Permit Access to the Instant Application by the Participating Offices
If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the World Intellectual Property Office (WIPO), and any other intellectual property offices in which a foreign application claiming priority to the instant patent application is filed access to the instant patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, WIPO, or other intellectual property office in which a foreign application claiming priority to the instant patent application is filed to have access to the instant patent application. In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the instant patent application with respect to: 1) the instant patent application-as-filed; 2) any foreign application to which the instant patent application claims priority under 35 U.S.C. 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the instant patent application; and 3) any U.S. application-as-filed from which benefit is sought in the instant patent application. In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing this Authorization.

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.		
Applicant 1		Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.		
Clear		
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117	☐ Joint Inventor
☐ Person to whom the inventor is obligated to assign.	☐ Person who shows sufficient proprietary interest	
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor :		
If the Applicant is an Organization check here. ☒		
Organization Name	International Business Machines Corporation	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Mailing Address Information:			
Address 1	New Orchard Road		
Address 2			
City	Armonk	State/Province	NY
Country i	US	Postal Code	10504
Phone Number		Fax Number	
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button. Add			

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Assignee 1				
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
Remove				
If the Assignee or Non-Applicant Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City		State/Province		
Country i		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add				

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Signature:

Remove

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications

Signature	/Amanda J. Smith/			Date (YYYY-MM-DD)	2015-06-03
First Name	Amanda J.	Last Name	Smith	Registration Number	71112

Additional Signature may be generated within this form by selecting the Add button.

Add

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☐ United States application or PCT international application number _____ filed on
_____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

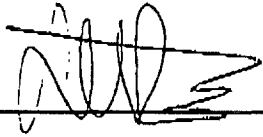
Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____



Date: 05/19/2015

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: _____ Date: _____

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☐ United States application or PCT international application number _____ filed on
_____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____ Date: _____

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: Nicolas L. Breil Date: 5/20/2015

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☐ United States application or PCT international application number _____ filed on
_____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

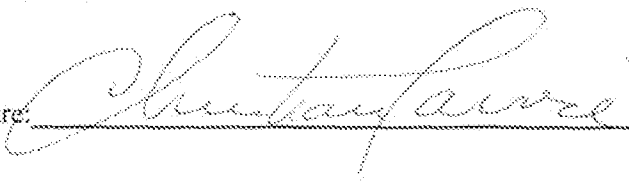
(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____ Date: _____

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: _____ Date: _____

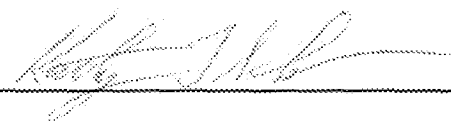
(3) Legal Name of Inventor: **Christian Lavoie**

Signature:  Date: May 13 2015

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature:  Date: May 13, 2015

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☐ United States application or PCT international application number _____ filed on
_____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____ Date: _____

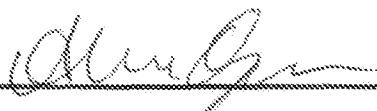
(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: _____ Date: _____

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature:  Date: 5/13/2015

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____



TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	
Filing Date	
First Named Inventor	Emre Alptekin et al.
Title	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
Art Unit	
Examiner Name	
Attorney Docket Number	FIS920150066US1

SIGNATURE of Applicant or Patent Practitioner

Signature	/Amanda J. Smith/	Date (Optional)	2015-06-03
Name	Amanda J. Smith	Registration Number	71,112
Title (if Applicant is a juristic entity)	Patent Practitioner for Applicant		
Applicant Name (if Applicant is a juristic entity)	International Business Machines Corporation		

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☐

*Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)



I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

45094

OR



I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:



The address associated with the above-mentioned Customer Number

OR



The address associated with Customer Number:

OR

Firm or Individual Name					
Address					
City		State		Zip	
Country					
Telephone			Email		

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

International Business Machines Corporation



Inventor or Joint Inventor (title not required below)



Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)



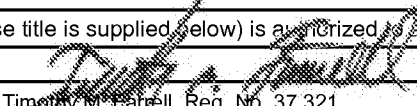
Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)



Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature		Date (Optional)	3-12-15
Name	Timothy M. Farrell, Reg. No. 37,321		
Title	Counsel		

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.



Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

PATENT ASSIGNMENT COVER SHEET

Electronic Version v1.1
Stylesheet Version v1.2

EPAS ID: PAT3380997

SUBMISSION TYPE:	NEW ASSIGNMENT
NATURE OF CONVEYANCE:	ASSIGNMENT
CONVEYING PARTY DATA	
Name	Execution Date
EMRE ALPTEKIN	05/19/2015
NICOLAS L. BREIL	05/20/2015
CHRISTIAN LAVOIE	05/13/2015
AHMET S. OZCAN	05/13/2015
KATHRYN T. SCHONENBERG	05/13/2015
RECEIVING PARTY DATA	
Name:	International Business Machines Corporation
Street Address:	New Orchard Road
City:	Armonk
State/Country:	NEW YORK
Postal Code:	10504
PROPERTY NUMBERS Total: 1	
Property Type	Number
Application Number:	14729298
CORRESPONDENCE DATA	
Fax Number:	(518)449-0047
<i>Correspondence will be sent to the e-mail address first; if that is unsuccessful, it will be sent using a fax number, if provided; if that is unsuccessful, it will be sent via US Mail.</i>	
Phone:	518-449-0044
Email:	ptocommunications@hoffmanwarnick.com, vfleming@hoffmanwarnick.com
Correspondent Name:	HOFFMAN WARNICK LLC
Address Line 1:	540 BROADWAY
Address Line 2:	4TH FLOOR
Address Line 4:	ALBANY, NEW YORK 12207
ATTORNEY DOCKET NUMBER:	FIS920150066US1
NAME OF SUBMITTER:	AMANDA J. SMITH
SIGNATURE:	/Amanda J. Smith/
DATE SIGNED:	06/03/2015
This document serves as an Oath/Declaration (37 CFR 1.63).	

Total Attachments: 8

source=FIS920150066US1_Filed_Dec-Assign#page1.tif

source=FIS920150066US1_Filed_Dec-Assign#page2.tif

source=FIS920150066US1_Filed_Dec-Assign#page3.tif

source=FIS920150066US1_Filed_Dec-Assign#page4.tif

source=FIS920150066US1_Filed_Dec-Assign#page5.tif

source=FIS920150066US1_Filed_Dec-Assign#page6.tif

source=FIS920150066US1_Filed_Dec-Assign#page7.tif

source=FIS920150066US1_Filed_Dec-Assign#page8.tif

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☒ United States application or PCT international application number 14/729,298 filed on
June 3, 2015.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

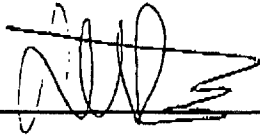
Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____



Date: 05/19/2015

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: _____ Date: _____

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☒ United States application or PCT international application number 14/729,298 filed on
June 3, 2015.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____ Date: _____

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: Nicolas L. Breil Date: 5/20/2015

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☒ United States application or PCT international application number 14/729,298 filed on
June 3, 2015.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

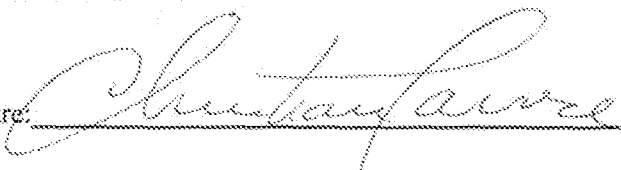
(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____ Date: _____

(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: _____ Date: _____

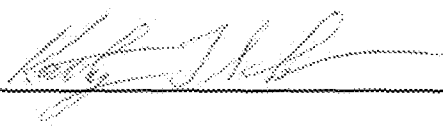
(3) Legal Name of Inventor: **Christian Lavoie**

Signature:  Date: May 13 2015

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature:  Date: May 13, 2015

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING
SAME**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☒ United States application or PCT international application number 14/729,298 filed on
June 3, 2015.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to

sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Emre Alptekin**

Signature: _____ Date: _____

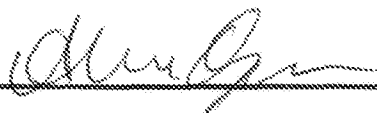
(2) Legal Name of Inventor: **Nicolas L. Breil**

Signature: _____ Date: _____

(3) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature:  Date: 5/13/2015

(5) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

SCORE Placeholder Sheet for IFW Content

Application Number: 14729298

Document Date: 06/03/2015

The presence of this form in the IFW record indicates that the following document type was received in electronic format on the date identified above. This content is stored in the SCORE database.

- Drawings – Other than Black and White Line Drawings

Since this was an electronic submission, there is no physical artifact folder, no artifact folder is recorded in PALM, and no paper documents or physical media exist. The TIFF images in the IFW record were created from the original documents that are stored in SCORE.

To access the documents in the SCORE database, refer to instructions below.

At the time of document entry (noted above):

- Examiners may access SCORE content via the eDAN interface.
- Other USPTO employees can bookmark the current SCORE URL (<http://Score.uspto.gov/ScoreAccessWeb/>).
- External customers may access SCORE content via the Public and Private PAIR interfaces.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/729,298

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	20 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	280
N/A	600
N/A	720
x 80 =	0.00
x 420 =	0.00
	0.00
	0.00
TOTAL	1600

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/729,298	06/03/2015	2812	1600	FIS920150066US1	20	3

CONFIRMATION NO. 2837

FILING RECEIPT



0000000075696482

45094
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

Date Mailed: 06/15/2015

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Emre Alptekin, Wappingers Falls, NY;
Nicolas L. Breil, San Jose, CA;
Christian Lavoie, Pleasantville, NY;
Ahmet S. Ozcan, Chappaqua, NY;
Kathryn T. Schonenberg, Wappingers Falls, NY;

Applicant(s)

International Business Machines Corporation, Armonk, NY;

Power of Attorney: The patent practitioners associated with Customer Number 45094

Domestic Applications for which benefit is claimed - None.

A proper domestic benefit claim must be provided in an Application Data Sheet in order to constitute a claim for domestic benefit. See 37 CFR 1.76 and 1.78.

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access - A proper **Authorization to Permit Access to Application by Participating Offices** (PTO/SB/39 or its equivalent) has been received by the USPTO.

If Required, Foreign Filing License Granted: 06/11/2015

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/729,298**

Projected Publication Date: 12/08/2016

Non-Publication Request: No

Early Publication Request: No

Title

CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

Preliminary Class

438

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	PATENT NUMBER	GROUP ART UNIT	FILE WRAPPER LOCATION
14/729,298		2812	e

PAIR Correspondence Address/Fee Address Change

The following fields have been changed to Customer Number 133964 on 07/17/2015 via Private PAIR in view of the certification copied below that authorized the change.

- Correspondence Address

The address for Customer Number 133964 is:

133964

HOFFMAN WARNICK LLC

540 Broadway

4th Floor

ALBANY, NY 12207

I certify, in accordance with 37 CFR 1.4(d)(4) that I am:

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature:	/Spencer K. Warnick/
Name:	Kyla Glickman
Registration Number:	40398



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	PATENT NUMBER	GROUP ART UNIT	FILE WRAPPER LOCATION
14/729,298		2812	e

PAIR Correspondence Address/Fee Address Change

The following fields have been changed to Customer Number 133964 on 07/17/2015 via Private PAIR in view of the certification copied below that authorized the change.

- Maintenance Fee Address

The address for Customer Number 133964 is:

133964

HOFFMAN WARNICK LLC

540 Broadway

4th Floor

ALBANY, NY 12207

I certify, in accordance with 37 CFR 1.4(d)(4) that I am:

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature:	/Spencer K. Warnick/
Name:	Kyla Glickman
Registration Number:	40398

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	14/729,298
Filing Date	06-03-2015
First Named Inventor	Emre Alptekin
Title	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
Art Unit	2812
Examiner Name	-
Attorney Docket Number	FIS920150066US1

SIGNATURE of Applicant or Patent Practitioner

Signature	/Spencer K. Warnick/	Date (Optional)	2016-02-18
Name	Spencer K. Warnick	Registration Number	40398
Title (if Applicant is a juristic entity)	Patent Practitioner for Applicant		
Applicant Name (if Applicant is a juristic entity)	GLOBALFOUNDRIES INC.		

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☐

*Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)

☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

OR

133964

☐ I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:

☒ The address associated with the above-mentioned Customer Number

OR

☐ The address associated with Customer Number:

OR

Firm or Individual Name					
Address					
City		State		Zip	
Country					
Telephone		Email			

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

GLOBALFOUNDRIES INC.

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature		Date (Optional)	
Name	David Cain		
Title	Senior Counsel, IP		

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ Total of forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

STATEMENT UNDER 37 CFR 3.73(c)Applicant/Patent Owner: GLOBALFOUNDRIES INC.Application No./Patent No.: 14/729,298 Filed/Issue Date: 06-03-2015Titled: CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAMEGLOBALFOUNDRIES INC., a Corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that, for the patent application/patent identified above, it is (choose one of options 1, 2, 3 or 4 below):

1. ☒ The assignee of the entire right, title, and interest.
2. ☐ An assignee of less than the entire right, title, and interest (check applicable box):
- ☐ The extent (by percentage) of its ownership interest is _____%. Additional Statement(s) by the owners holding the balance of the interest must be submitted to account for 100% of the ownership interest.
- ☐ There are unspecified percentages of ownership. The other parties, including inventors, who together own the entire right, title and interest are:

Additional Statement(s) by the owner(s) holding the balance of the interest must be submitted to account for the entire right, title, and interest.

3. ☐ The assignee of an undivided interest in the entirety (a complete assignment from one of the joint inventors was made). The other parties, including inventors, who together own the entire right, title, and interest are:

Additional Statement(s) by the owner(s) holding the balance of the interest must be submitted to account for the entire right, title, and interest.

4. ☐ The recipient, via a court proceeding or the like (e.g., bankruptcy, probate), of an undivided interest in the entirety (a complete transfer of ownership interest was made). The certified document(s) showing the transfer is attached.

The interest identified in option 1, 2 or 3 above (not option 4) is evidenced by either (choose one of options A or B below):

- A. ☐ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel _____, Frame _____, or for which a copy thereof is attached.
- B. ☒ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: Inventors To: International Business Machines CorporationThe document was recorded in the United States Patent and Trademark Office at
Reel 035779, Frame 0136, or for which a copy thereof is attached.2. From: International Business Machines Corporation To: GLOBALFOUNDRIES U.S. 2 LLCThe document was recorded in the United States Patent and Trademark Office at
Reel 037409, Frame 0869, or for which a copy thereof is attached.

[Page 1 of 2]

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

STATEMENT UNDER 37 CFR 3.73(c)3. From: GLOBALFOUNDRIES U.S. 2 LLC To: GLOBALFOUNDRIES INC.

The document was recorded in the United States Patent and Trademark Office at
 Reel 037542, Frame 0087, or for which a copy thereof is attached.

4. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

5. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

6. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
 Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

☒ As required by 37 CFR 3.73(c)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

/Spencer K. Warnick/

2016-02-18

Signature

Date

Spencer K. Warnick

40398

Printed or Typed Name

Title or Registration Number

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor 1					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Emre		Alptekin		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Wappingers Falls	State/Province	NY	Country of Residence i	US
Mailing Address of Inventor:					
Address 1	2070 Route 52				
Address 2					
City	Hopewell Junction		State/Province	NY	
Postal Code	12533	Country i	US		
Inventor 2					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Nicolas	L.	Breil		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	San Jose	State/Province	CA	Country of Residence i	US
Mailing Address of Inventor:					
Address 1	3401 Iron Point Drive				
Address 2					
City	San Jose		State/Province	CA	
Postal Code	95134	Country i	US		
Inventor 3					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Christian		Lavoie		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

City	Pleasantville	State/Province	NY	Country of Residence i	US
-------------	---------------	-----------------------	----	-------------------------------	----

Mailing Address of Inventor:

Address 1	1101 Kitchawan Road				
Address 2	Route 134				
City	Yorktown Heights	State/Province	NY		
Postal Code	10598	Country i	US		

Inventor 4	Remove
Legal Name	

Prefix	Given Name	Middle Name	Family Name	Suffix
	Ahmet	S.	Ozcan	
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				

City	Chappaqua	State/Province	NY	Country of Residence i	US
-------------	-----------	-----------------------	----	-------------------------------	----

Mailing Address of Inventor:

Address 1	2070 Route 52				
Address 2					
City	Hopewell Junction	State/Province	NY		
Postal Code	12533	Country i	US		

Inventor 5	Remove
Legal Name	

Prefix	Given Name	Middle Name	Family Name	Suffix
	Kathryn	T.	Schonenberg	
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				

City	Wappingers Falls	State/Province	NY	Country of Residence i	US
-------------	------------------	-----------------------	----	-------------------------------	----

Mailing Address of Inventor:

Address 1	1101 Kitchawan Road				
Address 2					
City	Yorktown Heights	State/Province	NY		
Postal Code	10598	Country i	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.					Add
---	--	--	--	--	---------------------

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).	
☐ An Address is being provided for the correspondence Information of this application.	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1	
		Application Number		
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME			
Customer Number	-45094-- <u>133964</u>			
Email Address			Add Email	Remove Email

Application Information:

Title of the Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME			
Attorney Docket Number	FIS920150066US1	Small Entity Status Claimed ☐		
Application Type	Nonprovisional			
Subject Matter	Utility			
Total Number of Drawing Sheets (if any)	5	Suggested Figure for Publication (if any)		

Filing By Reference :

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	-45094-- <u>133964</u>		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the application number blank.

Prior Application Status			Remove
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(d). When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(h)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

			Remove
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ^j (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Authorization to Permit Access:

☒ Authorization to Permit Access to the Instant Application by the Participating Offices

If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the World Intellectual Property Office (WIPO), and any other intellectual property offices in which a foreign application claiming priority to the instant patent application is filed access to the instant patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, WIPO, or other intellectual property office in which a foreign application claiming priority to the instant patent application is filed to have access to the instant patent application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the instant patent application with respect to: 1) the instant patent application-as-filed; 2) any foreign application to which the instant patent application claims priority under 35 U.S.C. 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the instant patent application; and 3) any U.S. application-as-filed from which benefit is sought in the instant patent application.

In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing this Authorization.

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1

Remove

If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.

Clear

☒ Assignee

☐ Legal Representative under 35 U.S.C. 117

☐ Joint Inventor

☐ Person to whom the inventor is obligated to assign.

☐ Person who shows sufficient proprietary interest

If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:

Name of the Deceased or Legally Incapacitated Inventor :

If the Applicant is an Organization check here. ☒

Organization Name

~~International Business Machines Corporation~~ -----

GLOBALFOUNDRIES INC.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Mailing Address Information:			
Address 1	-New Orchard Road- PO Box 309		
Address 2	Ugland House		
City	Armonk --- Grand Cayman	State/Province	NY--
Country i	US-- KY	Postal Code	10504-- KY1-1104
Phone Number		Fax Number	
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button. Add			

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Assignee 1				
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
				Remove
If the Assignee or Non-Applicant Assignee is an Organization check here.				☐
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City		State/Province		
Country i		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add				

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920150066US1
		Application Number	
Title of Invention	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME		

Signature:[Remove](#)

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications

Signature	/Spencer K. Warnick/		Date (YYYY-MM-DD)	2016-02-18	
First Name	Spencer K.	Last Name	Warnick	Registration Number	40398

Additional Signature may be generated within this form by selecting the Add button.

[Add](#)

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	24948822
Application Number:	14729298
International Application Number:	
Confirmation Number:	2837
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Spencer K. Warnick
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US1
Receipt Date:	18-FEB-2016
Filing Date:	03-JUN-2015
Time Stamp:	15:24:13
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	FIS920150066US1_POA_02-18-2016.pdf	215465 86d542383b608483d0a4aaf480e44f93bf82b87e	no	2

Warnings:

Information:

2	Assignee showing of ownership per 37 CFR 3.73	FIS920150066US1_373c_02-18-2016.pdf	122861 5f6a3968b87ddef13aaec184ec0f1be724609f4	no	3
Warnings:					
Information:					
3	Application Data Sheet	FIS920150066US1_CorrectedADS_02-18-2016.pdf	310625 5773a8366863d6f62610ce4e27b517b75199fc5a	no	8
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
Total Files Size (in bytes):			648951		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1

CONFIRMATION NO. 2837

POA ACCEPTANCE LETTER

133964
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207



Date Mailed: 03/03/2016

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 02/18/2016.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/mnguyen/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1

CONFIRMATION NO. 2837

POWER OF ATTORNEY NOTICE



133964
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

Date Mailed: 03/03/2016

NOTICE REGARDING CHANGE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 02/18/2016.

- The Power of Attorney to you in this application has been revoked by the applicant. Future correspondence will be mailed to the new address of record(37 CFR 1.33).

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/mnguyen/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/729,298	06/03/2015	2897	1600	FIS920150066US1	20	3

133964
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

CONFIRMATION NO. 2837
REPLACEMENT FILING RECEIPT



Date Mailed: 03/03/2016

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Emre Alptekin, Wappingers Falls, NY;
Nicolas L. Breil, San Jose, CA;
Christian Lavoie, Pleasantville, NY;
Ahmet S. Ozcan, Chappaqua, NY;
Kathryn T. Schonenberg, Wappingers Falls, NY;

Applicant(s)

GLOBALFOUNDRIES INC., Grand Cayman, CAYMAN ISLANDS;

Power of Attorney: The patent practitioners associated with Customer Number 133964

Domestic Applications for which benefit is claimed - None.

A proper domestic benefit claim must be provided in an Application Data Sheet in order to constitute a claim for domestic benefit. See 37 CFR 1.76 and 1.78.

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: No

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 06/11/2015

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/729,298**

Projected Publication Date: 12/08/2016

Non-Publication Request: No

Early Publication Request: No

Title

CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

Preliminary Class

257

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1	2837

133964 7590 04/21/2016
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

EXAMINER

NGUYEN, LAWRENCE-LIN T

ART UNIT	PAPER NUMBER
----------	--------------

2897

NOTIFICATION DATE	DELIVERY MODE
-------------------	---------------

04/21/2016

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PTOCommunications@hoffmanwarnick.com

Office Action Summary	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 2 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 02/26/2016.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☐ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-20 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☐ Claim(s) _____ is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☒ Claim(s) 1-20 are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☐ The drawing(s) filed on _____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☐ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date _____ | 4) ☐ Other: _____ |

DETAILED ACTION

Election/Restrictions

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. Restriction to one of the following inventions is required under 35 U.S.C. 121:
 - I. Claims **1-14**, drawn to **a method of**, classified in **438** subclass **648**.
 - II. Claims **15-20**, drawn to **a device of**, classified in **257** subclass **757**.
3. The inventions are independent or distinct, each from the other because of the following reasons:
 - a. Inventions Group I and Group II are related as process of making and product made. The inventions are distinct if either or both of the following can be shown: (1) that the process as claimed can be used to make another and materially different product or (2) that the product as claimed can be made by another and materially different process (MPEP § 806.05(f)).
 - b. In the instant case One aspect of the present invention involves depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate thereunder, e.g., source and drain. Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of

fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings.

4. **Applicant is advised that the reply to this requirement to be complete must include:**

(i) an election of a species to be examined even though the requirement may be traversed (37 CFR 1.143) **and**

(ii) identification of the claims encompassing the elected species or grouping of patentably indistinct species, including any claims subsequently added. An argument that a claim is allowable or that all claims are generic is considered nonresponsive unless accompanied by an election.

5. The election may be made with or without traverse. To preserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the election of species requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable on the elected species or grouping of patentably indistinct species.

6. Should applicant traverse on the ground that the species, or groupings of patentably indistinct species from which election is required, are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing them to be obvious variants or clearly admit on the record that this is the case. In either

Art Unit: 2897

instance, if the examiner finds one of the species unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103 or pre-AIA 35 U.S.C. 103(a) of the other species.

7. The examiner has required restriction between product or apparatus claims and process claims. Where applicant elects claims directed to the product/apparatus, and all product/apparatus claims are subsequently found allowable, withdrawn process claims that include all the limitations of the allowable product/apparatus claims should be considered for rejoinder. All claims directed to a nonelected process invention must include all the limitations of an allowable product/apparatus claim for that process invention to be rejoined.

8. In the event of rejoinder, the requirement for restriction between the product/apparatus claims and the rejoined process claims will be withdrawn, and the rejoined process claims will be fully examined for patentability in accordance with 37 CFR 1.104. Thus, to be allowable, the rejoined claims must meet all criteria for patentability including the requirements of 35 U.S.C. 101, 102, 103 and 112. Until all claims to the elected product/apparatus are found allowable, an otherwise proper restriction requirement between product/apparatus claims and process claims may be maintained. Withdrawn process claims that are not commensurate in scope with an allowable product/apparatus claim will not be rejoined. See MPEP § 821.04.

Additionally, in order for rejoinder to occur, applicant is advised that the process claims should be amended during prosecution to require the limitations of the product/apparatus claims. **Failure to do so may result in no rejoinder.** Further, note

Art Unit: 2897

that the prohibition against double patenting rejections of 35 U.S.C. 121 does not apply where the restriction requirement is withdrawn by the examiner before the patent issues. See MPEP § 804.01.

Conclusion

1. Any inquiry concerning this communication or earlier communications from the examiner should be directed to LAWRENCE-LINH T. NGUYEN whose telephone number is (571) 272-6267. The examiner can normally be reached on M-F (8AM - 4PM EST).
2. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Fernando Toledo can be reached on (571) 272-1867. The fax phone number for the organization where this application or proceeding is assigned is (571) 272-8300.
3. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at (866) 217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call (800) 786-9199 (in USA or CANADA) or (571) 272-1000.

/Lawrence-Linh T Nguyen/
Examiner, Art Unit 2897

/Fernando L. Toledo/
Supervisory Patent Examiner, Art
Unit 2897

Application/Control Number: 14/729,298
Art Unit: 2897

Page 7

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant:	Alptekin et al.	Conf. No.:	2837
Serial No.:	14/729,298	Art Unit:	2897
Filing Date:	06/03/2015	Examiner:	Nguyen, Lawrence-Lin T.
Title:	Contacts to Semiconductor Substrate and Methods of Forming Same	Docket No.:	FIS920150066US1 (GFFT-0376)

Mail Stop Amendment
Commissioner for Patents
PO Box 1450
Alexandria, VA 22313-1450

RESPONSE TO RESTRICTION REQUIREMENT

I. INTRODUCTORY COMMENTS

This paper is being filed in response to the Restriction Requirement dated 21 April 2016.

Applicant hereby elects invention I, claims 1-14, without traverse.

II. REMARKS

Claims 1–20 are pending in this application. The Office requires election between invention I (claims 1-14), drawn to “a method,” and invention II (claims 15-20), drawn to “a device.” (Office Action at page 2).

Applicant hereby elects invention I (claims 1-14) for examination on the merits, without traverse. Claims 15-20 stand withdrawn.

Should the Office require anything further from the Applicant, the Examiner is invited to contact Applicant’s undersigned representative at the number listed below.

Respectfully submitted,

/Amanda J. Smith /

Amanda J. Smith

Reg. No. 71,112

Date: June 20, 2016

Hoffman Warnick LLC
540 Broadway, 4th Floor
Albany, New York 12207
Phone: (518) 449-0044
Fax: (518) 449-0047

Electronic Acknowledgement Receipt

EFS ID:	26109411
Application Number:	14729298
International Application Number:	
Confirmation Number:	2837
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Amanda J Smith
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US1
Receipt Date:	20-JUN-2016
Filing Date:	03-JUN-2015
Time Stamp:	11:38:29
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Response to Election / Restriction Filed	FIS920150066US1_ResponseTo RestrictionRequirement.pdf	65761 6f75468b06dd8660e5d14d96353ba16274 344086	no	2

Warnings:

Information:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1	2837

133964 7590 09/26/2016
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

EXAMINER

NGUYEN, LAWRENCE-LIN T

ART UNIT	PAPER NUMBER
----------	--------------

2897

NOTIFICATION DATE	DELIVERY MODE
-------------------	---------------

09/26/2016

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PTOCommunications@hoffmanwarnick.com

Office Action Summary	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 05/20/2016.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☐ Claim(s) 1-20 is/are pending in the application.
5a) Of the above claim(s) 15-20 is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1-14 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☒ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 06/03/2015 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date <u>06/03/2015</u> | 4) ☐ Other: _____ |

Detailed Action

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. This non-final Office action is responsive to Applicants' application filed **06/20/2016**. Claims **1-20** are presented for examination and are rejected for the reasons indicated herein below.

Election/Restrictions

3. Applicant's election without traverse of Group I, claims **1-14** in the reply filed on **06/20/2016** is acknowledged.
4. Claims **15-20** are withdrawn pursuant to a restriction requirement.
5. Now claims **1-14** are pending.

Specification Objections

6. In the Specification, page 4, second paragraph, objected to because reference characters "**110**" and "**112**" have both been used to designate "**Substrate**". Appropriate correction is required.

Claim Rejections - 35 USC § 102(a)(1)

7. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

8. Claims **1-14** are rejected under 35 U.S.C. 102(a1), as being anticipated by **Heinrich et al., (US20120161324A1, hereinafter Heinrich)**.

9. In reclaim **1**, in FIG. 2, Heinrich discloses a method for forming a contact in a dielectric layer to a semiconductor substrate, the method compnsmg:

- forming a contact opening(**223**) in the dielectric layer(**221**) over the semiconductor substrate(**201**) to expose an upper portion of the semiconductor substrate(**201**);
- depositing a first liner layer(**205**) to conformally coat the contact opening(**223**);
- causing a portion of the first liner layer(**205**) to diffuse into the upper portion of the semiconductor substrate(**201**) to form a first intermix region(**253**) at the upper portion of the semiconductor substrate(**201**);
- depositing a refractory metal layer(**203**) over the first intermix

region(**YY**); and

- depositing a metal(**224a**) in the contact opening(**223**) thereby forming the contact(**223_Contact**) (See § **[0035-0046]**).

10. In reclaim **2**, in FIG. 2b, Heinrich discloses a method of claim **1**, further comprising: removing a remaining non-diffused portion of the first liner layer(**205**) to expose the first intermix region(**223_Bottom**) at the upper portion of the semiconductor substrate(**201**) prior to depositing the refractory metal layer (See § **[0039]**).

11. In reclaim **3**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein the depositing the first liner layer(**205**) includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb) (See § **[0006 0009]**).

12. In reclaim **4**, in FIG. XX, aaa discloses a method of claim **1**, wherein depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN) (See § **[0006]**).

13. In reclaim **5**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein depositing the first liner layer includes performing physical vapor deposition (See § **[0006]**).

14. In reclaim **7**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal(**224a**) in the opening includes

Art Unit: 2897

depositing tungsten (W) (See § [0045]).

15. In reclaim 8, in FIG. 2, Heinrich discloses a method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:

- forming a contact opening(223) in the dielectric layer(221) to expose an upper-portion of each of the source(252_Source) and the drain(252_Drain);
- depositing a first liner layer(205) to conformally coat each contact opening(223) and causing a portion of the first liner layer(205) to diffuse into the upper portion of each of the source(252_Source) and the drain(252_Drain) to form a first intermix region(253) at the upper portion of the each of the source(252_Source) and the drain(252_Drain);
- depositing a refractory metal layer(203) over the first intermix region(253); and
- depositing a metal(224a) to substantially coat the refractory metal layer(203) and substantially fill each of the openings(223) thereby forming the set of contacts(223_Contact) (See § [0035-0046]).

16. In reclaim 9, in FIG. 2b, Heinrich discloses a method of claim 8, further comprising: removing a remaining non-diffused portion of the first liner layer(205) to expose the intermix region(223_Bottom) prior to depositing the refractory metal layer(203) (See § [0039]).

Art Unit: 2897

17. In reclaim **10**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb) (*See § [0006 0009]*).

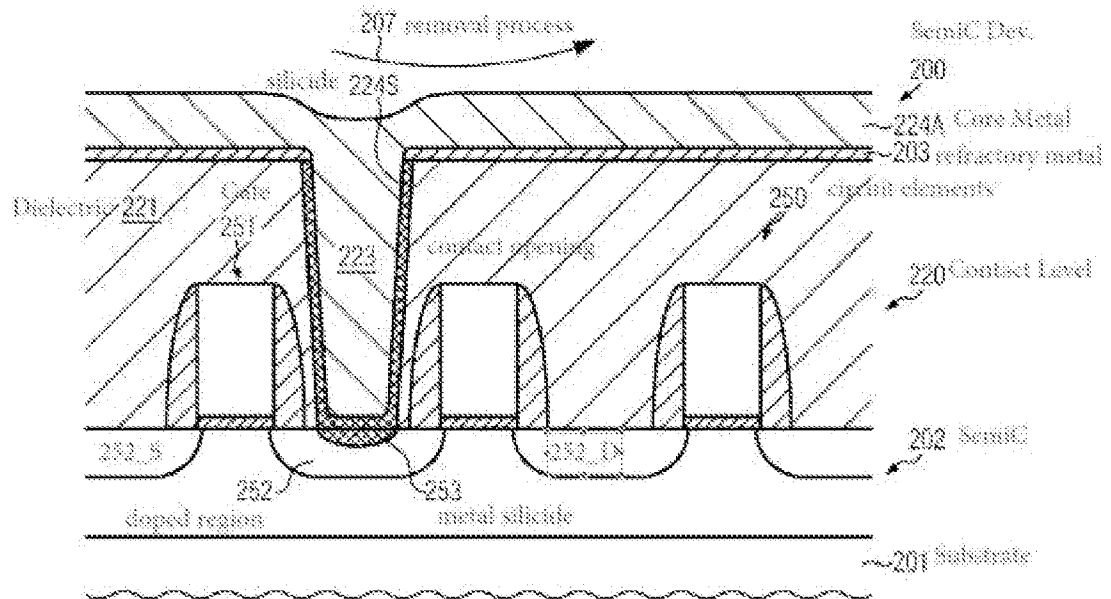
18. In reclaim **11**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN) (*See § [0006]*).

12. In reclaim **12**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the first liner layer(**205**) includes performing physical vapor deposition (*See § [0006]*).

19. In reclaim **14**, in FIG. 2, Heinrich discloses a method of claim 13, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal(**224a**) in the opening includes depositing tungsten (W) (*See § [0045]*).

Art Unit: 2897

20. (Also see US20120161324A1_FIG2hMod which is reproduced below for convenience).



Claim Rejections - 35 USC § 103

22. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

A patent may not be obtained through the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in Graham v. John Deere Co., 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103(a) are summarized as follows: (See MPEP Ch. 2141)

Determining the scope and contents of the prior art;

Ascertaining the differences between the prior art and the claims in issue; Resolving the level of ordinary skill in the pertinent art; and Evaluating evidence of secondary considerations for indicating obviousness or nonobviousness

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

23. Claims **6 & 13** is rejected under 35 U.S.C. 103(a) as being unpatentable by **Heinrich et al.**, (**US20120161324A1**, hereinafter **Heinrich**) in view of **Hu et al.**, (**US20140027883A1**, hereinafter **Hu**).

24. In reclaim **6**, Heinrich is silent in the teaching of a method of claim 1, further comprising: polishing the metal; and performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor

Art Unit: 2897

substrate and the upper surface of semiconductor substrate, wherein the second intermix region is composed of materials used for the first intermix region and a material used for the refractory metal layer.

25. But in the same field of endeavor of invention, Hu teaches a method of claim 1, further comprising: polishing the metal; and performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and the upper surface of semiconductor substrate, wherein the second intermix region is composed of materials used for the first intermix region and a material used for the refractory metal layer.

26. Hu is evidence that ordinary workers skilled in the art would find reasons, suggestions or motivations to modify the device of Hu with Heinrich. Therefore, at the time the invention was made, it would have been obvious to use the teaching of Hu in the device of Heinrich since the superlattice formed from the first refractory metal nitride material and the second refractory metal nitride material may also improve the thermal stability of the bottom electrode or interconnect (*See § [0016]*).

27. In reclaim **13**, in FIG. 2, Heinrich in view of Hu, Hu teaches a method of claim 8, further comprising: polishing the metal in each of the openings; and performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials used for the first intermix region and a material used for the refractory metal layer (*See § [0016 0027]*).

Conclusion

29. Any inquiry concerning this communication or earlier communications from the examiner should be directed to LAWRENCE-LINH T. NGUYEN whose telephone number is (571) 272-6267. The examiner can be reached on M-F (8AM - 4PM EST).

30. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Fernando Toledo can be reached on (571) 272-1867. The fax phone number for the organization where this application or proceeding is assigned is (571) 272-8300.

31. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at (866) 217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call (800) 786-9199 (in USA or CANADA) or (571) 272-1000.

/Lawrence-Linh T Nguyen/

Examiner, Art Unit 2897

/Fernando L. Toledo/

Supervisory Patent Examiner, Art
Unit 2897

Application/Control Number: 14/729,298
Art Unit: 2897

Page 11

Notice of References Cited	Application/Control No. 14/729,298		Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN		Art Unit 2897	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2012/0161324 A1	06-2012	Heinrich; Jens	H01L21/28518	257/757
*	B	US-2014/0027883 A1	01-2014	Hu; Yongjun Jeff	H01L21/743	257/534
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 14729298	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

CPC- SEARCHED		
Symbol	Date	Examiner
H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261.	09/14/2016	LTN

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
Each of the CPC classes listed are searched in combination and/or delimited with text. See EAST search for details	09/14/2016	LTN

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

/LAWRENCE-LINH T NGUYEN/ Examiner.Art Unit 2897	
--	--

Receipt date: 06/03/2015

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

14729298 - GAI: 2897

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number			
	Filing Date			
	First Named Inventor	Emre Alptekin et al.		
	Art Unit			
	Examiner Name			
	Attorney Docket Number	FIS920150066US1		

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	8278200	B2	2012-10-02	Lavoie et al.	

If you wish to add additional U.S. Patent citation information please click the Add button.

Add

U.S.PATENT APPLICATION PUBLICATIONS						Remove
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1					

If you wish to add additional U.S. Published Application citation information please click the Add button.

Add

FOREIGN PATENT DOCUMENTS								Remove
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐

If you wish to add additional Foreign Patent Document citation information please click the Add button

Add

NON-PATENT LITERATURE DOCUMENTS				Remove
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T5	

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /L.N./

Receipt date: 06/03/2015 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14729298 - GAU: 2897
	Filing Date		
	First Named Inventor	Emre Alptekin et al.	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	FIS920150066US1	

1	Filed Application titled "Metal Semiconductor Alloy Contact Resistance Improvement," US Application No. 14/172,525 filed February 4, 2014, Attorney Docket No. FIS920130170US1, 37 pages.	☐
---	---	--------------------------

If you wish to add additional non-patent literature document citation information please click the Add button **Add**

EXAMINER SIGNATURE

Examiner Signature	/Lawrence-Linh Nguyen/	Date Considered	09/13/2016
--------------------	------------------------	-----------------	------------

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

Receipt date: 06/03/2015 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14729298 - GAU: 2897
	Filing Date		
	First Named Inventor	Emre Alptekin et al.	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	FIS920150066US1	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Amanda J. Smith/	Date (YYYY-MM-DD)	2015-06-03
Name/Print	Amanda J. Smith	Registration Number	71,112

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
		"1817" and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:42
S1	0	(14/729298).APP.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2016/09/12 20:02
S2	48	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S3	367	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S4	104	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03
S5	133	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03
S6	104	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:21
S7	9310	("GLOBALFOUNDRIES INC."	US-PGPUB;	OR	ON	2016/09/12

		GLOBALFOUNDRIES).AS.	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			20:22
S8	55	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 21:07
S9	181	"7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:21
S10	26	(US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US-20130009217- \$ or US-20090289302-\$ or US- 20140145319-\$ or US-20140138837-\$ or US-20090302480-\$ or US-20100078771- \$ or US-20050085064-\$ or US- 20060246656-\$ or US-20120132966-\$) did.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:22
S11	13120	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:24
S12	23052	S1 S2 S3 S4 S5 S6 S7 S11 S9 S10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:25
S13	1273	S12 and ((opening trench via hole) with dielectric) and ((liner or barrier) with metal) and (metal with (contact or fill))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:25
S14	3241	S12 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:25

S15	92	S12 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:26
S16	516	S13 S14	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:27
S17	28	S16 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:27
S18	11581	((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:44
S19	650	S12 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:45
S20	110	S12 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:47
S21	22	S19 S20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:48
S22	170152	((Gate Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:52
S23	63156	S22 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/09/12 22:55

			DERWENT; IBM_TDB			
S24	22129	S22 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S25	1183	S22 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S26	2141	S22 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 23:09
S27	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S28	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S29	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S30	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S31	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S32	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB;	OR	ON	2016/09/13

			USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			11:03
S33	245794	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:05
S34	13131	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:12
S35	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US-20130009217- \$ or US-20090289302-\$ or US- 20140145319-\$ or US-20140138837-\$ or US-20090302480-\$ or US-20100078771- \$ or US-20050085064-\$ or US- 20060246656-\$ or US-20120132966-\$) .did.("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:13
S36	259027	S27 S28 S29 S30 S31 S32 S33 S34 S35	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:15
S37	10581	S36 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT;	AND	ON	2016/09/13 13:25

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S38	29272	S36 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:25
S39	261	S36 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S40	356	S36 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S41	6783	S36 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:29
S42	8119	S37 S38	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S43	2155	S37 S38 S41	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S44	6	S39 S40	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S45	0	S43 S44	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29

S46	43	S43 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:42
S47	9	S43 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:43
S48	1709	S43 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 16:49
S49	78	S43 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "AlzO3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:11
S50	43406	S36 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:30

		processing metalorganic) adj (CVD)))				
S51	1853	S43 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:31
S52	1481	S48 S51	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:32
S53	62	S52 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "AlzO3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S54	5	S52 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S55	29	S52 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS;	OR	ON	2016/09/13 17:32

			EPO; JPO; DERWENT; IBM_TDB			
S56	34	S52 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Cobalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S57	1481	S52 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S58	1452	S52 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolot\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S59	318	S52 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolot\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:40
S60	318	S57 S58 S59	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:40

9/ 13/ 2016 9:36:40 PM
C:\ Users\ Inguyen16\ Documents\ EAST\ Workspaces\ 14729298.wsp



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
 United States Patent and Trademark Office
 Address: COMMISSIONER FOR PATENTS
 P.O. Box 1450
 Alexandria, Virginia 22313-1450
 www.uspto.gov

BIB DATA SHEET

CONFIRMATION NO. 2837

SERIAL NUMBER 14/729,298	FILING or 371(c) DATE 06/03/2015 RULE	CLASS 257	GROUP ART UNIT 2897	ATTORNEY DOCKET NO. FIS920150066US1	
APPLICANTS GLOBALFOUNDRIES INC., Grand Cayman, CAYMAN ISLANDS; INVENTORS Emre Alptekin, Wappingers Falls, NY; Nicolas L. Breil, San Jose, CA; Christian Lavoie, Pleasantville, NY; Ahmet S. Ozcan, Chappaqua, NY; Kathryn T. Schonenberg, Wappingers Falls, NY;					
** CONTINUING DATA ***** ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 06/11/2015					
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /LAWRENCE-LINH T NGUYEN/ Acknowledged Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY NY	SHEETS DRAWINGS 5	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 3
ADDRESS HOFFMAN WARNICK LLC 540 Broadway 4th Floor ALBANY, NY 12207 UNITED STATES					
TITLE CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME					
FILING FEE RECEIVED 1600	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

<i>Index of Claims</i> 	Application/Control No. 14729298	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	09/13/2016								
	1	✓								
	2	✓								
	3	✓								
	4	✓								
	5	✓								
	6	✓								
	7	✓								
	8	✓								
	9	✓								
	10	✓								
	11	✓								
	12	✓								
	13	✓								
	14	✓								
	15	N								
	16	N								
	17	N								
	18	N								
	19	N								
	20	N								



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1

CONFIRMATION NO. 2837

PUBLICATION NOTICE



OC000000087752371

133964
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

Title:CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

Publication No.US-2016-0358860-A1

Publication Date:12/08/2016

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Office of Public Records. The Office of Public Records can be reached by telephone at (703) 308-9726 or (800) 972-6382, by facsimile at (703) 305-8759, by mail addressed to the United States Patent and Trademark Office, Office of Public Records, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <http://pair.uspto.gov/>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant:	Alptekin et al.	Conf. No.:	2837
Serial No.:	14/729,298	Art Unit:	2897
Filing Date:	06/03/2015	Examiner:	Nguyen, Lawrence-Lin T.
Title:	Contacts to Semiconductor Substrate and Methods of Forming Same	Docket No.:	FIS920150066US1 (GFFT-0376)

Mail Stop Amendment
Commissioner for Patents
PO Box 1450
Alexandria, VA 22313-1450

AMENDMENT

I. INTRODUCTORY COMMENTS

This paper is being filed in response to the Office Action of September 26, 2016. Please amend the above-referenced patent application as follows:

Electronic Acknowledgement Receipt

EFS ID:	27736255
Application Number:	14729298
International Application Number:	
Confirmation Number:	2837
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Amanda J Smith
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US1
Receipt Date:	08-DEC-2016
Filing Date:	03-JUN-2015
Time Stamp:	15:10:36
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		FIS920150066US1_ResponseTo OfficeAction_12-08-16.pdf	59140	yes	15
			7e3b2a103cfaab781b33709f2f9a62ef9fac04 824		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Applicant Arguments/Remarks Made in an Amendment	9	15
	Specification	6	8
	Claims	2	5
	Amendment/Req. Reconsideration-After Non-Final Reject	1	1

Warnings:

Information:

Total Files Size (in bytes):	59140
-------------------------------------	-------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IV. REMARKS

Claims 1-14 are currently pending. In this Amendment, claims 2, 6, and 13 and the Specification have been amended, and claims 15-20 have been cancelled without prejudice. No new matter is added. Reconsideration and allowance are requested in view of the accompanying amendments and the remarks below. Applicant is not conceding in this application that the amended or cancelled claims are not patentable over references cited by the Examiner, as the present claim amendments and cancellation are for facilitating expeditious allowance of the claimed subject matter. Further, Applicant reserves the right to pursue the full scope of the subject matter of the original claims in a subsequent patent application that claims priority to the instant application.

Specification Objections

The Specification has been objected to for alleged informalities. Specifically, the Examiner alleges that characters “110” and “112” have both been used to designate “substrate”. (Office Action, page 2). Applicant has amended the Specification to address the alleged informalities. As such, Applicant respectfully requests withdrawal of the objection to the Specification.

Rejection under 35 U.S.C. § 102(a)(1)

In the Office Action, claims 1-14 are rejected under 35 U.S.C. § 102(a)(1), as being anticipated by Heinrich et al. (US Publication 20120161324A1), hereinafter referred to as “Heinrich”. Applicant respectfully traverses the rejections for at least the reasons that follow.

With respect to claim 1, the Examiner alleges that Heinrich teaches “causing a portion of the first liner layer (205) to diffuse into the upper portion of the semiconductor substrate (201) to form a first intermix region (253) at the upper portion of the semiconductor substrate (201).” (Office Action, page 3). Applicant respectfully disagrees. Heinrich’s discloses “during the process 207, the heavily doped semiconductor region 252 may also be exposed within the contact opening 223, while on the other hand corresponding sidewall spacers 205S may be formed on inner surface areas 223S of the contact opening.” (Heinrich, paragraph [0039]). That is, Heinrich’s liner 205 is removed from an upper surface of the heavily doped semiconductor region 252. Heinrich goes on to disclose that “a layer of a refractory metal species 203 may be formed...[and] a heat treatment or a sequence of heat treatments 204 may be applied so as to initiate a chemical reaction of the layer 203 with any underlying semiconductor materials” to form metal silicide 253. (Heinrich, paragraph [0040]). That is, Heinrich’s metal silicide 253, which is equated to Applicant’s first intermix region, is not formed by causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate as claimed. Rather, Heinrich’s liner 205 is removed prior to the forming of Heinrich’s metal silicide 253, and thus, Heinrich’s metal silicide 253 is formed from Heinrich’s refractory metal species 203 and Heinrich’s highly doped semiconductor region 252 by heat treatment.

In another embodiment, Heinrich discloses “the refractory metal 203 may be deposited directly on the semiconductor layer 205 without a preceding material removal process. Consequently, upon initiating a silicidation process, metal silicide may be efficiently formed with the contact opening 223 and in and above the doped region 252, as is also previously explained, while also a metal silicide may be formed on the layer 205 above the dielectric material 221.” (Heinrich, paragraph [0044]). That is, in this embodiment, Heinrich’s metal silicide 253 is

formed above layer 205. Thus, Heinrich does not disclose “causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate” as claimed.

In light of the foregoing, Applicant respectfully submits that Heinrich fails to disclose each and every feature of Applicant’s claim 1. As such, Applicant respectfully requests withdrawal of the 35 U.S.C. § 102(a)(1) rejection of claim 1.

With respect to claim 8, Applicant submits that claim 8 recites *inter alia*, “causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain.” As should be clear from the above discussion relative to claim 1, Heinrich does not disclose at least this feature. As such, Applicant respectfully requests withdrawal of the 35 U.S.C. § 102(a)(1) rejection of claim 8.

Should the Office decide to issue a subsequent action issuing any new rejection of claims 1 and/or 8 in this Application, Applicants request that the subsequent action be made non-final, as the new rejection would not be necessitated by Applicants’ amendments herein. (MPEP 706.07(a), “Under present practice, second or any subsequent actions on the merits shall be final, except where the examiner introduces a new ground of rejection that is neither necessitated by applicant's amendment of the claims nor based on information submitted in an information disclosure statement filed during the period set forth in 37 CFR 1.97(c) with the fee set forth in 37 CFR 1.17(p)”).

Claims 2-5, 7, 9-12, and 14 are allowable due to their dependency from allowable claims 1 and 8, respectively. As such, Applicant respectfully requests withdrawal of the 35 U.S.C. § 102(a)(1) rejection of claims 2-5, 7, 9-12, and 14. With respect to claims 6 and 13, Applicant

submits that the Examiner fails to individually address claims 6 and 13 even though he rejects claims 1-14 under 35 U.S.C. § 102(a)(1). As such, Applicant does not yet bear a burden to address any 35 U.S.C. § 102(a)(1) of claims 6 and 13.

More specifically, with respect to claims 2 and 9, since Heinrich fails to disclose the forming of the first intermix region by causing diffusion of the first liner layer, it stands that Heinrich cannot stand to disclose removing a remaining non-diffused portion of the first liner layer to expose the intermix region as claimed. As previously discussed, the Examiner equates Heinrich's metal silicide 253 to the claimed intermix region and Heinrich's liner 205 to the claimed first liner layer. However, Heinrich's liner 205 is removed prior to the formation of Heinrich's metal silicide 253, so it is unclear how the removal of Heinrich's liner 205 exposes Heinrich's metal silicide 253.

With respect to claims 4, 7, 11, and 14, Applicant submits that while Heinrich discloses "layer 203 may comprise any appropriate metal species, such as nickel, platinum and the like, while in other cases cobalt may be used" (Heinrich, paragraph [0040]), this disclosure cannot stand to teach the process of wherein "the depositing the refractory metal layer includes depositing a *bilayer* of titanium and titanium nitride" as claimed. (*Emphasis added*). Should the Examiner persist in this assertion, Applicant respectfully requests that the Examiner provide support for his conclusion that Heinrich teaches the claimed "depositing of a bilayer".

Rejection under 35 U.S.C. § 103

In the Office Action, claims 6 and 13 are rejected under 35 U.S.C. § 103(a) as being unpatentable over Heinrich, hereinafter referred to as "Heinrich", in view of Hu et al. (US

Publication 20140027883A1), hereinafter referred to as “Hu”. Applicant respectfully traverses the rejection for at least the reasons that follow.

First, Applicant notes that in the rejection of claims 6 and 13 under 35 U.S.C. § 103(a), the Examiner merely reproduces the elements of the claims and provides a blanket allegation that Heinrich and Hu teach these features without pointing to specific teachings of Heinrich and/or Hu for which the Examiner relies on in rejection of the claims.

Under MPEP § 2142, “[t]he examiner bears the initial burden of factually supporting any prima facie conclusion of obviousness. If the examiner does not produce a prima facie case, the applicant is under no obligation to submit evidence of nonobviousness.” “The Federal Circuit has stated that ‘rejections on obviousness cannot be sustained with mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.’ In re Kahn, 441 F.3d 977, 988, 78 USPQ2d 1329, 1336 (Fed. Cir. 2006). See also KSR, 550 U.S. at ___, 82 USPQ2d at 1396 (quoting Federal Circuit statement with approval).” (MPEP § 2142.) As such, Applicant does not yet bear the burden to address the 35 U.S.C. § 103(a) rejections of claims 6 and 13.

However, in the interest of expeditious prosecution, Applicant submits that neither Heinrich nor Hu teach at least “performing a laser anneal of the opening such that a second intermix region is formed at the upper surface of the semiconductor substrate...[and] the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.” Heinrich only teaches forming one metal silicide region 253 within contact opening 223 which is equated to Applicant’s first intermix region. Heinrich does not teach the forming a second intermix region as claimed. Hu teaches “a metal-insulator-metal

(MIM) capacitor having a bottom electrode formed from at least two portions of a metal nitride material.” (Hu, paragraph [0002]).

V. CONCLUSION

Applicant respectfully submits that the Application as presented is in condition for allowance. Should the Examiner believe that anything further is necessary in order to place the application in better condition for allowance, the Examiner is requested to contact Applicant's undersigned representative at the telephone number listed below.

Respectfully submitted,

/Amanda J. Smith /

Amanda J. Smith

Reg. No. 71,112

Date: December 8, 2016

Hoffman Warnick LLC
540 Broadway, 4th Floor
Albany, New York 12207
Phone: (518) 449-0044
Fax: (518) 449-0047

III. AMENDMENTS TO THE SPECIFICATION

Please replace paragraphs [0011], [0013], and [0014] with the following amended paragraphs [0011], [0013], and [0014], respectively:

[0011] Referring now to FIG. 1, a semiconductor structure 100 is shown including a semiconductor substrate 110, an isolation region 112, and a gate structure 114 within a dielectric layer 120 over substrate 112. Substrate ~~[[112]]~~110 may be employed as a semiconductor layer of a semiconductor-on-insulator substrate (not shown), or a bulk substrate (not shown) as understood in the art. Substrate ~~[[112]]~~110 may include any now known or later developed semiconductor material including but not limited to silicon, germanium, silicon germanium, silicon carbide, and those consisting essentially of one or more III-V compound semiconductors having a composition defined by the formula $Al_{X1}Ga_{X2}In_{X3}As_{Y1}P_{Y2}N_{Y3}Sb_{Y4}$, where $X1, X2, X3, Y1, Y2, Y3,$ and $Y4$ represent relative proportions, each greater than or equal to zero and $X1+X2+X3+Y1+Y2+Y3+Y4=1$ (1 being the total relative mole quantity). Other suitable substrates include II-VI compound semiconductors having a composition $Zn_{A1}Cd_{A2}Se_{B1}Te_{B2}$, where $A1, A2, B1,$ and $B2$ are relative proportions each greater than or equal to zero and $A1+A2+B1+B2=1$ (1 being a total mole quantity). Furthermore, a portion or entire substrate may be strained. Substrate ~~[[112]]~~110 may be doped, undoped or contain doped and undoped regions therein.

[0013] Further, FIG. 1 illustrates that a gate 114 may be formed on substrate ~~[[112]]~~110. Gate structure 114 may include a gate dielectric material 116 and a gate conductor 118. Gate structure

114 may be used to control output current of the semiconducting device through electrical or magnetic fields. Gate dielectric 116 may include, but is not limited to: HfO_2 , ZrO_2 , La_2O_3 , Al_2O_3 , TiO_2 , SrTiO_3 , LaAlO_3 , Y_2O_3 , HfO_xN_y , ZrO_xN_y , $\text{La}_2\text{O}_x\text{N}_y$, $\text{Al}_2\text{O}_x\text{N}_y$, TiO_xN_y , SrTiO_xN_y , LaAlO_xN_y , $\text{Y}_2\text{O}_x\text{N}_y$, SiON , SiN_x , a silicate thereof, and an alloy thereof. Each value of x is independently from 0.5 to 3 and each value of y is independently from 0 to 2. In some embodiments, a multilayered gate dielectric structure comprising different gate dielectric materials, e.g., silicon dioxide, and a dielectric metal oxide can be formed and used as gate dielectric material 116. Gate conductor 118 may include, but is not limited to: doped polysilicon, doped silicon germanium, an elemental metal (e.g., tungsten, titanium, tantalum, aluminum, nickel, ruthenium, palladium and platinum), an alloy of at least two elemental metals, an elemental metal nitride (e.g., tungsten nitride, aluminum nitride, and titanium nitride), or multilayered combinations thereof.

[0014] Gate structure 114 may be formed by depositing gate dielectric 116 on substrate [[112]]110. As used herein, “depositing” may include any now known or later developed techniques appropriate for the material to be deposited including but are not limited to, for example: chemical vapor deposition (CVD), low-pressure CVD (LPCVD), plasma-enhanced CVD (PECVD), semi-atmosphere CVD (SACVD) and high density plasma CVD (HDPCVD), rapid thermal CVD (RTCVD), ultra-high vacuum CVD (UHVCVD), limited reaction processing CVD (LRPCVD), metalorganic CVD (MOCVD), sputtering deposition, ion beam deposition, electron beam deposition, laser assisted deposition, thermal oxidation, thermal nitridation, spin-on methods, physical vapor deposition (PVD), atomic layer deposition (ALD), chemical oxidation, molecular beam epitaxy (MBE), plating, evaporation. Additionally, subsequent to the

deposition of gate dielectric 116, gate conductor 118 can be deposited on gate dielectric 116. The precise thickness of gate dielectric 116 and gate conductor 118 may vary widely with the intended application.

II. AMENDMENTS TO THE CLAIMS

The following listing of claims replaces all prior versions, and listings, of claims in the application:

1. (Original) A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;

depositing a first liner layer to conformally coat the contact opening;

causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;

depositing a refractory metal layer over the first intermix region; and

depositing a metal in the contact opening thereby forming the contact.

2. (Currently Amended) The method of claim 1, further comprising:

removing a remaining non-diffused portion of the first liner layer to expose the first intermix region at the upper portion of the semiconductor ~~substrate prior~~ substrate prior to depositing the refractory metal layer.

3. (Original) The method of claim 1, wherein the depositing the first liner layer includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb).

4. (Original) The method of claim 1, wherein depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

5. (Original) The method of claim 1, wherein depositing the first liner layer includes performing physical vapor deposition.

6. (Currently Amended) The method of claim 1, further comprising:

polishing the metal; and

performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and ~~the upper surface of semiconductor substrate~~ a bottom surface of the contact, wherein the second intermix region is composed of materials ~~used for~~ of the first intermix region and a material used for the refractory metal layer.

7. (Original) The method of claim 1, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

8. (Original) A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:

forming a contact opening in the dielectric layer to expose an upper portion of each of the source and the drain;

depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of each of the source and the drain;

depositing a refractory metal layer over the first intermix region; and

depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.

9. (Original) The method of claim 8, further comprising:

removing a remaining non-diffused portion of the first liner layer to expose the intermix region prior to depositing the refractory metal layer.

10. (Original) The method of claim 8, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb)..

11. (Original) The method of claim 8, wherein the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

12. (Original) The method of claim 8, wherein the depositing the first liner layer includes performing physical vapor deposition.

13. (Currently Amended) The method of claim 8, further comprising:

polishing the metal in each of the openings; and

performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials ~~used for~~ of the first intermix region and a material used for the refractory metal layer.

14. (Original) The method of claim 13, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

15-20. (Cancelled)

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/729,298	Filing Date 06/03/2015	☐ To be Mailed
---	---	----------------------------------	---------------------------------------

 ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

AMENDMENT	(Column 1)	(Column 2)	(Column 3)	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	12/09/2016	CLAIMS REMAINING AFTER AMENDMENT					
	Total (37 CFR 1.16(i))	* 14	Minus	** 20	= 0	X \$80 =	0
	Independent (37 CFR 1.16(h))	* 2	Minus	***3	= 0	X \$420 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE						0	

AMENDMENT	(Column 1)	(Column 2)	(Column 3)	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
		CLAIMS REMAINING AFTER AMENDMENT					
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE							

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

 LIE
STELLA LITTLE

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1	2837

133964 7590 05/08/2017
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

EXAMINER

NGUYEN, LAWRENCE-LIN T

ART UNIT	PAPER NUMBER
----------	--------------

2897

NOTIFICATION DATE	DELIVERY MODE
-------------------	---------------

05/08/2017

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PTOCommunications@hoffmanwarnick.com

Office Action Summary	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 12/08/2016.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
- 2a) ☒ This action is **FINAL**. 2b) ☐ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-20 is/are pending in the application.
5a) Of the above claim(s) 15-20 is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☒ Claim(s) 1-14 is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☐ The drawing(s) filed on ____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. ____. |
| 2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date ____. | 4) ☐ Other: ____. |

Detailed Action

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. This **Final Office** action is responsive to Applicants' Amendment/Req.

Reconsideration-After-Non-Final Rejection filed on **12/08/2016**.

a. In response to the Applicants' request, claims **1-20** are presented for examination: Claims **15-20** are canceled and claims **2, 6, & 13** are amended.

The amended claims are supported by the original disclosure of the present application and therefore, no new matter/issue is raised.

b. Now claims **1-14** are pending.

3. For applicant's benefit portions of the cited reference(s) have been cited to aid in the review of the rejection(s). While every attempt has been made to be thorough and consistent within the rejection it is noted that the *PRIOR ART MUST BE CONSIDERED IN ITS ENTIRETY, INCLUDING DISCLOSURES THAT TEACH AWAY FROM THE CLAIMS*. See MPEP 2141.02 VI.

Specification Objections

4. In the Specification objection to the reference characters "**110**" and "**112**" is withdrawn.

Claim Rejections - 35 USC § 102(a)(1)

5. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

6. Claims **1-14** are rejected under 35 U.S.C. 102(a1), as being anticipated by

Heinrich et al., (US20120161324A1, hereinafter Heinrich).

7. In reclaim **1**, in FIG. 2, Heinrich discloses a method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

- forming a contact opening(**223**) in the dielectric layer(**221**) over the semiconductor substrate(**201**) to expose an upper portion of the semiconductor substrate(**201**);
 - depositing a first liner layer(**205**) to conformally coat the contact opening(**223**);
 - causing a portion of the first liner layer(**205**) to diffuse into the upper portion of the semiconductor substrate(**201**) to form a first intermix region(**253**) at the upper portion of the semiconductor substrate(**201**);
 - depositing a refractory metal layer(**203**) over the first intermix region;
- and

- depositing a metal(**224a**) in the contact opening(**223**) thereby forming the contact(**223_Contact**) (See § **[0035-0046]**).
8. In reclaim **2**, in FIG. 2b, Heinrich discloses a method of claim **1**, further comprising: removing a remaining non-diffused portion of the first liner layer(**205**) to expose the first intermix region(**223_Bottom**) at the upper portion of the semiconductor substrate(**201**) prior to depositing the refractory metal layer (See § **[0039]**).
9. In reclaim **3**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein the depositing the first liner layer(**205**) includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb) (See § **[0006 0009]**).
10. In reclaim **4**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN) (See § **[0006]**).
11. In reclaim **5**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein depositing the first liner layer includes performing physical vapor deposition (See § **[0006]**).
12. In reclaim **7**, in FIG. 2, Heinrich discloses a method of claim **1**, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal(**224a**) in the opening includes depositing tungsten (W) (See § **[0045]**).

Art Unit: 2897

13. In reclaim 8, in FIG. 2, Heinrich discloses a method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:

- forming a contact opening(223) in the dielectric layer(221) to expose an upper-portion of each of the source(252_Source) and the drain(252_Drain);
- depositing a first liner layer(205) to conformally coat each contact opening(223) and causing a portion of the first liner layer(205) to diffuse into the upper portion of each of the source(252_Source) and the drain(252_Drain) to form a first intermix region(253) at the upper portion of the each of the source(252_Source) and the drain(252_Drain);
- depositing a refractory metal layer(203) over the first intermix region(253); and
- depositing a metal(224a) to substantially coat the refractory metal layer(203) and substantially fill each of the openings(223) thereby forming the set of contacts(223_Contact) (See § [0035-0046]).

14. In reclaim 9, in FIG. 2b, Heinrich discloses a method of claim 8, further comprising: removing a remaining non-diffused portion of the first liner layer(205) to expose the intermix region(223_Bottom) prior to depositing the refractory metal layer(203) (See § [0039]).

Art Unit: 2897

15. In reclaim **10**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb) (*See § [0006 0009]*).

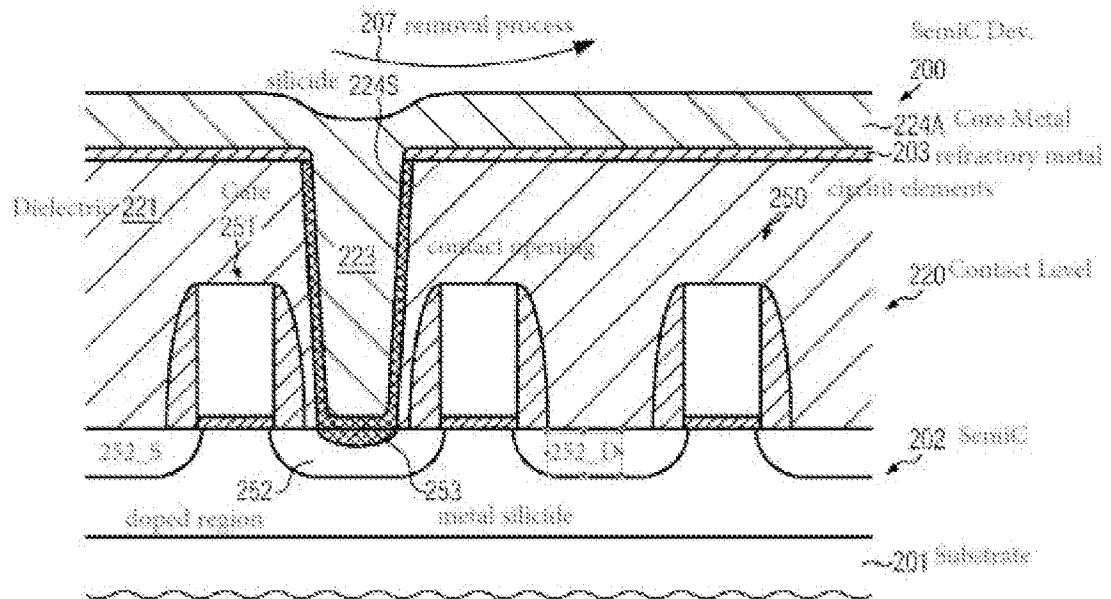
16. In reclaim **11**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN) (*See § [0006]*).

17. In reclaim **12**, in FIG. 2, Heinrich discloses a method of claim 8, wherein the depositing the first liner layer(**205**) includes performing physical vapor deposition (*See § [0006]*).

18. In reclaim **14**, in FIG. 2, Heinrich discloses a method of claim 13, wherein the depositing the first liner layer(**205**) includes depositing nickel platinum (NiPt), the depositing the refractory metal layer(**203**) includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal(**224a**) in the opening includes depositing tungsten (W) (*See § [0045]*).

Art Unit: 2897

2. (Also see US20120161324A1_FIG2hMod which is reproduced below for convenience).



Claim Rejections - 35 USC § 103

19. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

A patent may not be obtained through the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in Graham v. John Deere Co., 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103(a) are summarized as follows: (See MPEP Ch. 2141)

Determining the scope and contents of the prior art;

Ascertaining the differences between the prior art and the claims in issue; Resolving the level of ordinary skill in the pertinent art; and Evaluating evidence of secondary considerations for indicating obviousness or nonobviousness

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

20. Claims **6 & 13** is rejected under 35 U.S.C. 103(a) as being unpatentable by

Heinrich et al., (**US20120161324A1**, hereinafter **Heinrich**) in view of **Hu et al.**,

(**US20140027883A1**, hereinafter **Hu**).

21. In reclaim **6**, Heinrich is silent in the teaching of a method of claim 1, further comprising: polishing the metal; and performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and the upper surface of semiconductor substrate, wherein the second

Art Unit: 2897

intermix region is composed of materials used for the first intermix region and a material used for the refractory metal layer.

22. But in the same field of endeavor of invention, Hu teaches a method of claim 1, further comprising: polishing the metal; and performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and a bottom surface of the contact, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.

23. Hu is evidence that ordinary workers skilled in the art would find reasons, suggestions or motivations to modify the device of Hu with Heinrich. Therefore, at the time the invention was made, it would have been obvious to use the teaching of Hu in the device of Heinrich since the superlattice formed from the first refractory metal nitride material and the second refractory metal nitride material may also improve the thermal stability of the bottom electrode or interconnect (*See § [0016]*).

24. In reclaim **13**, in FIG. 2, Heinrich in view of Hu, Hu teaches a method of claim 8, further comprising: polishing the metal in each of the openings; and performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer (*See § [0016 0027]*).

Response to Applicant's Remarks

25. With respect to the claims, the following are addressed:
26. Applicant's arguments filed with respect to the claims **1-14**, under USC 102/103 have been fully considered and they are addressed in this Final office action.

Conclusion

27. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

28. A shortened statutory period for reply to this final action is set to expire THREE(3) MONTHS from the mailing date of this action. In the event a first reply is filed within TWO(2) MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE(3)-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX(6) MONTHS from the mailing date of this final action.

29. Any inquiry concerning this communication or earlier communications from the examiner should be directed to **LAWRENCE-LINH T. NGUYEN** (Email address **Lawrence-Linh.Nguyen@uspto.gov**) whose telephone number is (571) 272-6267.

The examiner can be reached on M-F (8AM - 4PM EST).

30. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Fernando Toledo can be reached on (571) 272-1867. The fax phone number for the organization where this application or proceeding is assigned is (571) 272-8300.

Art Unit: 2897

31. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at (866) 217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call (800) 786-9199 (in USA or CANADA) or (571) 272-1000.

/Lawrence-Linh T Nguyen/

Examiner, Art Unit 2897

/Fernando L. Toledo/

Supervisory Patent Examiner, Art
Unit 2897

Notice of References Cited	Application/Control No. 14/729,298		Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN		Art Unit 2897	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2012/0161324 A1	06-2012	Heinrich; Jens	H01L21/28518	257/757
*	B	US-2014/0027883 A1	01-2014	Hu; Yongjun Jeff	H01L21/743	257/534
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	1	(14/729298).APP.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2017/04/17 13:15
L4	0	("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.).PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2017/04/17 13:32
L5	0	("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.).PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2017/04/17 13:32
L6	0	("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.).PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2017/04/17 13:33
L7	53	("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/17 13:33
S1	0	(14/729298).APP.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	OFF	2016/09/12 20:02

			DERWENT; IBM_TDB			
S2	48	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S3	367	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S4	104	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03
S5	133	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03
S6	104	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:21
S7	9310	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES).AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:22
S8	55	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 21:07
S9	181	"7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:21

		"20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502"				
S10	26	(US-20130240990-\$ or US-20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US-20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US-20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US-20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:22
S11	13120	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:24
S12	23052	S1 S2 S3 S4 S5 S6 S7 S11 S9 S10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:25
S13	1273	S12 and ((opening trench via hole) with dielectric) and ((liner or barrier) with metal) and (metal with (contact or fill))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:25
S14	3241	S12 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:25
S15	92	S12 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:26
S16	516	S13 S14	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:27
S17	28	S16 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:27

S18	11581	((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:44
S19	650	S12 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:45
S20	110	S12 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:47
S21	22	S19 S20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:48
S22	170152	((Gate Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:52
S23	63156	S22 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:55
S24	22129	S22 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S25	1183	S22 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S26	2141	S22 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/09/12 23:09

		Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	DERWENT; IBM_TDB			
S27	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S28	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S29	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S30	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S31	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S32	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S33	245794	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:05
S34	13131	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:12
S35	276	("8278200").PN. ("20070249149"	US-PGPUB;	OR	ON	2016/09/13

		"20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			11:13
S36	259027	S27 S28 S29 S30 S31 S32 S33 S34 S35	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:15
S37	10581	S36 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:25
S38	29272	S36 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:25
S39	261	S36 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28

S40	356	S36 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S41	6783	S36 and ((Fusing Fused Fussion Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:29
S42	8119	S37 S38	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S43	2155	S37 S38 S41	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S44	6	S39 S40	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S45	0	S43 S44	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S46	43	S43 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:42
S47	9	S43 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:43
S48	1709	S43 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/09/13 16:49

			DERWENT; IBM_TDB			
S49	78	S43 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "Al2O3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:11
S50	43406	S36 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:30
S51	1853	S43 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:31

		"thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))				
S52	1481	S48 S51	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:32
S53	62	S52 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "AlzO3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S54	5	S52 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S55	29	S52 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S57	1481	S52 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S58	1452	S52 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39

S59	318	S52 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:40
S60	318	S57 S58 S59	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:40
S61	16	("20010035558" "20050124128" "20080171411" "20080272357" "20080303068" "20100075474" "6087209" "6093243" "6440828" "6812082" "7259110" "7736964" "7755114" "7785947").PN. OR ("8304819").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:07
S62	21	("20040056281" "20040121590" "20050020093" "20050026380" "20060141712" "20060166494" "20070042580" "20070141798" "20090134497" "20110084320" "20120202326" "20120248550" "20140015104" "6323519" "6730570" "7087515" "7220668" "7348283" "7679192" "8304819" "8492217").PN. OR ("9252019").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:15
S63	2	"20130049219"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:43
S64	21	("20040056281" "20040121590" "20050020093" "20050026380" "20060141712" "20060166494" "20070042580" "20070141798" "20090134497" "20110084320" "20120202326" "20120248550" "20140015104" "6323519" "6730570" "7087515" "7220668" "7348283" "7679192" "8304819" "8492217").PN. OR ("9252019").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:44
S65	13	("20040067658" "20040150075" "20050110153" "20060003577" "6878620" "7087515" "7199044" "7253097" "7253524" "7309658" "7466027").PN. OR ("7679192").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:49
S66	4	"20130049219" "20070152334"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:51
S67	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S68	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT;	OR	ON	2016/09/14 15:09

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S69	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S70	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S71	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S72	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S73	245794	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S74	13131	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S75	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09

		"20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.				
S76	259027	S67 S68 S69 S70 S71 S72 S73 S74 S75	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S77	10581	S76 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 15:09
S78	29272	S76 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S79	6783	S76 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S80	2155	S77 S78 S79	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 15:09
S81	1709	S80 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2016/09/14 15:09

			IBM_TDB			
S82	1853	S80 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S83	1481	S81 S82	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 15:09
S84	34	S83 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S85	8119	S77 S78	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 17:08
S86	2155	S77 S78 S79	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 17:08
S87	407	S76 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused"	US-PGPUB; USPAT;	OR	ON	2016/09/14 17:35

		((("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) with (Remov\$3 Extract\$3 Etch\$3) with (Source Drain))	USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S88	45	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" ((("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) with (Remov\$3 Extract\$3 Etch\$3) with (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 17:37
S89	0	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" ((("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 20:05
S90	118	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" ((("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 20:06
S91	33	S90 and ((Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 20:06
S92	1480	S80 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:52
S93	22	(US-20150221740-\$ or US-20130240990-\$ or US-20100078771-\$ or US-20090023261-\$ or US-20120161324-\$ or US-20130119483-\$ or US-20140319620-\$ or US-20140246698-\$ or US-20120104498-\$ or US-20080040697-\$ or US-20060163631-\$ or US-20070122956-\$ or US-20130049219-\$ or US-20070152334-\$).did. or (US-8304819-\$ or US-9224675-\$ or US-8742510-\$ or US-9093425-\$ or US-9252019-\$ or US-6440828-\$ or US-7679192-\$).did. or (US-20110084320-\$).did.	US-PGPUB; USPAT; DERWENT	OR	ON	2016/09/14 21:52
S94	16	S93 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:52

S95	4	S93 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:57
S96	22455	H01L21/7684 H01L21/76841 H01L21/76843 H01L21/76846 H01L21/7685 H01L21/76879 H01L21/76883 H01L23/481 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 10:32
S97	69	"20150187896" "5565708"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 10:33
S98	639	S96 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:44
S99	21	S98 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:44
S100	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S101	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S102	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S103	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/09/15 12:45

			DERWENT; IBM_TDB			
S104	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S105	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S106	245815	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S107	13151	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S108	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45

		"20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.				
S109	259067	S100 S101 S102 S103 S104 S105 S106 S107 S108	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S110	19	S109 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:57
S111	20386	S109 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 13:52
S112	4016	S109 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 13:53
S113	0	S98 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 13:54
S114	0	S112 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:00
S115	84	S109 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:01
S116	0	S98 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:01
S117	34	("20020019107" "20020102810"	US-PGPUB;	OR	ON	2016/09/15

		"20040206620" "20040259308" "20050087879" "20050263890" "20070145525" "20080073708" "20080138948" "20080188055" "20080272421" "20100001325" "20100025853" "20100187655" "20100213571" "20110057292" "20130011990" "5981331" "6219125" "6511879" "6593616" "6746916" "6753618" "6881642" "6906420" "7071054" "7268072" "7304367" "7329939" "7368343" "7545009" "7772678" "8105891").PN. OR ("8835274").URPN.	USPAT; USOCR			14:04
S118	2	"20140027883"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/15 14:04
S119	0	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:11
S120	0	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) with ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:12
S121	36	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:12

4/ 17/ 2017 2:10:16 PM
C:\ Users\ Inguyen16\ Documents\ EAST\ Workspaces\ 14729298.F.wsp

III. AMENDMENTS TO THE SPECIFICATION

Please replace paragraphs [0011], [0013], and [0014] with the following amended paragraphs [0011], [0013], and [0014], respectively:

[0011] Referring now to FIG. 1, a semiconductor structure 100 is shown including a semiconductor substrate 110, an isolation region 112, and a gate structure 114 within a dielectric layer 120 over substrate 112. Substrate ~~[[112]]~~110 may be employed as a semiconductor layer of a semiconductor-on-insulator substrate (not shown), or a bulk substrate (not shown) as understood in the art. Substrate ~~[[112]]~~110 may include any now known or later developed semiconductor material including but not limited to silicon, germanium, silicon germanium, silicon carbide, and those consisting essentially of one or more III-V compound semiconductors having a composition defined by the formula $\text{Al}_{\text{X1}}\text{Ga}_{\text{X2}}\text{In}_{\text{X3}}\text{As}_{\text{Y1}}\text{P}_{\text{Y2}}\text{N}_{\text{Y3}}\text{Sb}_{\text{Y4}}$, where X1, X2, X3, Y1, Y2, Y3, and Y4 represent relative proportions, each greater than or equal to zero and $\text{X1}+\text{X2}+\text{X3}+\text{Y1}+\text{Y2}+\text{Y3}+\text{Y4}=1$ (1 being the total relative mole quantity). Other suitable substrates include II-VI compound semiconductors having a composition $\text{Zn}_{\text{A1}}\text{Cd}_{\text{A2}}\text{Se}_{\text{B1}}\text{Te}_{\text{B2}}$, where A1, A2, B1, and B2 are relative proportions each greater than or equal to zero and $\text{A1}+\text{A2}+\text{B1}+\text{B2}=1$ (1 being a total mole quantity). Furthermore, a portion or entire substrate may be strained. Substrate ~~[[112]]~~110 may be doped, undoped or contain doped and undoped regions therein.

[0013] Further, FIG. 1 illustrates that a gate 114 may be formed on substrate ~~[[112]]~~110. Gate structure 114 may include a gate dielectric material 116 and a gate conductor 118. Gate structure

114 may be used to control output current of the semiconducting device through electrical or magnetic fields. Gate dielectric 116 may include, but is not limited to: HfO_2 , ZrO_2 , La_2O_3 , Al_2O_3 , TiO_2 , SrTiO_3 , LaAlO_3 , Y_2O_3 , HfO_xN_y , ZrO_xN_y , $\text{La}_2\text{O}_x\text{N}_y$, $\text{Al}_2\text{O}_x\text{N}_y$, TiO_xN_y , SrTiO_xN_y , LaAlO_xN_y , $\text{Y}_2\text{O}_x\text{N}_y$, SiON , SiN_x , a silicate thereof, and an alloy thereof. Each value of x is independently from 0.5 to 3 and each value of y is independently from 0 to 2. In some embodiments, a multilayered gate dielectric structure comprising different gate dielectric materials, e.g., silicon dioxide, and a dielectric metal oxide can be formed and used as gate dielectric material 116. Gate conductor 118 may include, but is not limited to: doped polysilicon, doped silicon germanium, an elemental metal (e.g., tungsten, titanium, tantalum, aluminum, nickel, ruthenium, palladium and platinum), an alloy of at least two elemental metals, an elemental metal nitride (e.g., tungsten nitride, aluminum nitride, and titanium nitride), or multilayered combinations thereof.

[0014] Gate structure 114 may be formed by depositing gate dielectric 116 on substrate [[112]]110. As used herein, “depositing” may include any now known or later developed techniques appropriate for the material to be deposited including but are not limited to, for example: chemical vapor deposition (CVD), low-pressure CVD (LPCVD), plasma-enhanced CVD (PECVD), semi-atmosphere CVD (SACVD) and high density plasma CVD (HDPCVD), rapid thermal CVD (RTCVD), ultra-high vacuum CVD (UHVCVD), limited reaction processing CVD (LRPCVD), metalorganic CVD (MOCVD), sputtering deposition, ion beam deposition, electron beam deposition, laser assisted deposition, thermal oxidation, thermal nitridation, spin-on methods, physical vapor deposition (PVD), atomic layer deposition (ALD), chemical oxidation, molecular beam epitaxy (MBE), plating, evaporation. Additionally, subsequent to the

deposition of gate dielectric 116, gate conductor 118 can be deposited on gate dielectric 116. The precise thickness of gate dielectric 116 and gate conductor 118 may vary widely with the intended application.

Search Notes 	Application/Control No. 14729298	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

CPC- SEARCHED		
Symbol	Date	Examiner
H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261.	09/14/2016	LTN

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
Each of the CPC classes listed are searched in combination and/or delimited with text. See EAST search for details	09/14/2016	LTN

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

--	--

<i>Index of Claims</i> 	Application/Control No. 14729298	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE							
Final	Original	09/13/2016	04/17/2017						
	1	✓	✓						
	2	✓	✓						
	3	✓	✓						
	4	✓	✓						
	5	✓	✓						
	6	✓	✓						
	7	✓	✓						
	8	✓	✓						
	9	✓	✓						
	10	✓	✓						
	11	✓	✓						
	12	✓	✓						
	13	✓	✓						
	14	✓	✓						
	15	N	N						
	16	N	N						
	17	N	N						
	18	N	N						
	19	N	N						
	20	N	N						

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant(s): Emre Alptekin

Conf. No.: 2837

Serial No.: 14/729,298

Art. Unit: 2897

Filed: June 3, 2015

Examiner: Nguyen, Lawrence –Lin T.

Docket No.: FIS920150066US1
(GFFT-0376-US)

Title: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME**

Mail Stop AF
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AFTER-FINAL RESPONSE

Sir:

I. INTRODUCTORY COMMENTS

This paper is being filed in response to the Final Office Action dated May 8, 2017.

Please reconsider the above-identified patent application in light of the following:

II. LISTING OF THE CLAIMS

The following listing of the claims is provided merely as a courtesy. No claims have been amended herein.

1. (Original) A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;

depositing a first liner layer to conformally coat the contact opening;

causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;

depositing a refractory metal layer over the first intermix region; and

depositing a metal in the contact opening thereby forming the contact.

2. (Previously Presented) The method of claim 1, further comprising:

removing a remaining non-diffused portion of the first liner layer to expose the first intermix region at the upper portion of the semiconductor substrate prior to depositing the refractory metal layer.

3. (Original) The method of claim 1, wherein the depositing the first liner layer includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb).

4. (Original) The method of claim 1, wherein depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

5. (Original) The method of claim 1, wherein depositing the first liner layer includes performing physical vapor deposition.

6. (Previously Presented) The method of claim 1, further comprising:

polishing the metal; and

performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and a bottom surface of the contact, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.

7. (Original) The method of claim 1, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

8. (Original) A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:

forming a contact opening in the dielectric layer to expose an upper -portion of each of the source and the drain;

depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain;

depositing a refractory metal layer over the first intermix region; and

depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.

9. (Original) The method of claim 8, further comprising:

removing a remaining non-diffused portion of the first liner layer to expose the intermix region prior to depositing the refractory metal layer.

10. (Original) The method of claim 8, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb)..

11. (Original) The method of claim 8, wherein the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

12. (Original) The method of claim 8, wherein the depositing the first liner layer includes performing physical vapor deposition.

13. (Previously Presented) The method of claim 8, further comprising:

polishing the metal in each of the openings; and

performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.

14. (Original) The method of claim 13, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

15-20. (Cancelled)

III. APPLICANT-INITATED INTERVIEW SUMMARY

Initially, Applicant would like to thank the Examiner for conducting the telephone interview with Applicant's representative, Amanda J. Smith (Reg. No. 71,112) on June 30, 2017. During the Interview, Applicant and the Examiner discussed the 35 U.S.C. § 102 rejection of claim 1. Applicant's representative argued that the rejection was flawed as Heinrich does not show each and every feature of Applicant's claim 1. The Examiner agreed with such arguments. Further, the Examiner acknowledged his failure to address Applicant's arguments that were submitted in the Amendment filed on December 8, 2016. At the conclusion of the Interview, the Examiner requested that Applicant file a response including an Interview Summary evidencing the agreements. Applicant has reproduced the arguments herein that were previously submitted on December 8, 2016. In addition, the Examiner indicated that he would, if necessary, issue a new non-final Office Action in response to the instant response.

IV. REMARKS

Claims 1-14 are pending in this application. No claims have been amended. Claims 15-20 were previously cancelled. Applicant reserves the right to pursue the full scope of the subject matter of the original claims in a subsequent patent application that claims priority to the instant application. Reconsideration in view of the following remarks is respectfully requested.

Rejections under 35 U.S.C. § 102(a)(1)

Claims 1-14 are rejected under 35 U.S.C. § 102(a)(1), as allegedly being anticipated by Heinrich et al., (US 20120161324A1, hereinafter Heinrich). Applicant respectfully traverses the rejections for at least the reasons that follow.

With respect to claim 1, the Examiner alleges that Heinrich teaches “causing a portion of the first liner layer (205) to diffuse into the upper portion of the semiconductor substrate (201) to form a first intermix region (253) at the upper portion of the semiconductor substrate (201).” (Office Action, page 3). Applicant respectfully disagrees. Heinrich’s discloses “during the process 207, the heavily doped semiconductor region 252 may also be exposed within the contact opening 223, while on the other hand corresponding sidewall spacers 205S may be formed on inner surface areas 223S of the contact opening.” (Heinrich, paragraph [0039]). That is, Heinrich’s liner 205 is removed from an upper surface of the heavily doped semiconductor region 252. Heinrich goes on to disclose that “a layer of a refractory metal species 203 may be formed...[and] a heat treatment or a sequence of heat treatments 204 may be applied so as to initiate a chemical reaction of the layer 203 with any underlying semiconductor materials” to form metal silicide 253. (Heinrich, paragraph [0040]). That is, Heinrich’s metal silicide 253, which is equated to Applicant’s first intermix region, is not formed by causing a portion of the

first liner layer to diffuse into the upper portion of the semiconductor substrate as claimed.

Rather, Heinrich's liner 205 is removed prior to the forming of Heinrich's metal silicide 253, and thus, Heinrich's metal silicide 253 is formed from Heinrich's refractory metal species 203 and Heinrich's highly doped semiconductor region 252 by heat treatment.

In another embodiment, Heinrich discloses "the refractory metal 203 may be deposited directly on the semiconductor layer 205 without a preceding material removal process. Consequently, upon initiating a silicidation process, metal silicide may be efficiently formed with the contact opening 223 and in and above the doped region 252, as is also previously explained, while also a metal silicide may be formed on the layer 205 above the dielectric material 221." (Heinrich, paragraph [0044]). That is, in this embodiment, Heinrich's metal silicide 253 is formed above layer 205. Thus, Heinrich does not disclose "causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate" as claimed.

In light of the foregoing, Applicant respectfully submits that Heinrich fails to disclose each and every feature of Applicant's claim 1. As such, Applicant respectfully requests withdrawal of the 35 U.S.C. § 102(a)(1) rejection of claim 1.

With respect to claim 8, Applicant submits that claim 8 recites, *inter alia*, "causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain." As should be clear from the above discussion relative to claim 1, Heinrich does not disclose at least this feature. As such, Applicant respectfully requests withdrawal of the 35 U.S.C. § 102(a)(1) rejection of claim 8.

Should the Office decide to issue a subsequent action issuing any new rejection of claims 1 and/or 8 in this Application, Applicant requests that the subsequent action be made non-final, as the new rejection would not be necessitated by Applicant's amendments herein. MPEP 706.07(a), "Under present practice, second or any subsequent actions on the merits shall be final, except where the examiner introduces a new ground of rejection that is neither necessitated by applicant's amendment of the claims nor based on information submitted in an information disclosure statement filed during the period set forth in 37 CFR 1.97(c) with the fee set forth in 37 CFR 1.17(p)".

Claims 2-5, 7, 9-12, and 14 are allowable due to their dependency from allowable claims 1 and 8, respectively. As such, Applicant respectfully requests withdrawal of the 35 U.S.C. § 102(a)(1) rejection of claims 2-5, 7, 9-12, and 14. With respect to claims 6 and 13, Applicant submits that the Examiner fails to individually address claims 6 and 13 even though he rejects claims 1-14 under 35 U.S.C. § 102(a)(1). As such, Applicant does not yet bear a burden to address any 35 U.S.C. § 102(a)(1) of claims 6 and 13.

More specifically, with respect to claims 2 and 9, since Heinrich fails to disclose the forming of the first intermix region by causing diffusion of the first liner layer, it stands that Heinrich cannot stand to disclose removing a remaining non-diffused portion of the first liner layer to expose the intermix region as claimed. As previously discussed, the Examiner equates Heinrich's metal silicide 253 to the claimed intermix region and Heinrich's liner 205 to the claimed first liner layer. However, Heinrich's liner 205 is removed prior to the formation of Heinrich's metal silicide 253, so it is unclear how the removal of Heinrich's liner 205 exposes Heinrich's metal silicide 253.

With respect to claims 4, 7, 11, and 14, Applicant submits that while Heinrich discloses “layer 203 may comprise any appropriate metal species, such as nickel, platinum and the like, while in other cases cobalt may be used” (Heinrich, paragraph [0040]), this disclosure does not teach that “depositing the refractory metal layer includes depositing a *bilayer* of titanium and titanium nitride” as claimed. (*Emphasis added*). Should the Examiner persist in this assertion, Applicant respectfully requests that the Examiner provide support for his conclusion that Heinrich teaches the claimed “depositing of a bilayer”.

Rejections under 35 U.S.C. § 103

Claims 6 and 13 are rejected under 35 U.S.C. § 103(a) as allegedly being unpatentable by Heinrich in view of Hu et al., (US20140027883A1, hereinafter Hu). Applicant respectfully traverses the rejection for at least the reasons that follow.

Claims 6 and 13 are each allowable due to their dependency from allowable claims 1 and 8, respectively as well as for their own additionally claimed subject matter. Hu does not remedy the deficiencies discussed above relative to Heinrich, nor is it alleged to do so. Further, Applicant submits that neither Heinrich nor Hu teach at least “performing a laser anneal of the opening such that a second intermix region is formed at the upper surface of the semiconductor substrate...[and] the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.” Heinrich only teaches forming one metal silicide region 253 within contact opening 223 which is equated to Applicant’s first intermix region. Heinrich does not teach the forming a second intermix region as claimed. Hu teaches “a metal-insulator-metal (MIM) capacitor having a bottom electrode formed from at least two portions of a metal nitride material.” (Hu, paragraph [0002]).

As a result, Applicant respectfully requests withdrawal of the 35 U.S.C. § 103 rejections of claims 6 and 13.

V. CONCLUSION

Applicant respectfully submits that the Application as presented is in condition for allowance. Should the Examiner believe that anything further is necessary in order to place the application in better condition for allowance, the Examiner is requested to contact Applicant's undersigned representative at the telephone number listed below.

Respectfully submitted,

/ Amanda J. Smith /
Amanda J. Smith
Reg. No. 71,112

Date: July 7, 2017

Hoffman Warnick LLC
540 Broadway, 4th Floor
Albany, New York 12207
Phone: (518) 449-0044
Fax: (518) 449-0047

Electronic Acknowledgement Receipt

EFS ID:	29714598
Application Number:	14729298
International Application Number:	
Confirmation Number:	2837
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Amanda J Smith
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US1
Receipt Date:	07-JUL-2017
Filing Date:	03-JUN-2015
Time Stamp:	12:31:06
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		FIS920150066US1_Amendmen tToFinalOfficeAction1_07-07-1 7.pdf	52804 3ab308798479f2f24a861c2d0ec5b3dfdfcbb 0e2	yes	11

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Response After Final Action	1	1
	Claims	2	5
	Applicant summary of interview with examiner	6	6
	Applicant Arguments/Remarks Made in an Amendment	7	11

Warnings:

Information:

Total Files Size (in bytes):	52804
-------------------------------------	-------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/729,298	Filing Date 06/03/2015	☐ To be Mailed
---	---	----------------------------------	---------------------------------------

 ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

FOR	NUMBER FILED (Column 1)	NUMBER EXTRA (Column 2)	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

AMENDMENT	(Column 1)	(Column 2)	(Column 3)	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	07/07/2017	CLAIMS REMAINING AFTER AMENDMENT					
	Total (37 CFR 1.16(i))	* 14	Minus	** 20	= 0	X \$80 =	0
	Independent (37 CFR 1.16(h))	* 2	Minus	***3	= 0	X \$420 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE							0

AMENDMENT	(Column 1)	(Column 2)	(Column 3)	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
		CLAIMS REMAINING AFTER AMENDMENT					
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE							

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

 LIE
 SULONDA D. STEVENSON

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

133964 7590 10/19/2017
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

EXAMINER

NGUYEN, LAWRENCE-LIN T

ART UNIT

PAPER NUMBER

2897

DATE MAILED: 10/19/2017

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

14/729,298

06/03/2015

Emre Alptekin

FIS920150066US1

2837

TITLE OF INVENTION: CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	01/19/2018

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

133964 7590 10/19/2017
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1	2837

TITLE OF INVENTION: CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	01/19/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
NGUYEN, LAWRENCE-LIN T	2897	257-757000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted: ☐ Issue Fee ☐ Publication Fee (No small entity discount permitted) ☐ Advance Order - # of Copies _____	4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above) ☐ A check is enclosed. ☐ Payment by credit card. Form PTO-2038 is attached. ☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).
---	---

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1	2837

133964 7590 10/19/2017
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

EXAMINER

NGUYEN, LAWRENCE-LIN T

ART UNIT	PAPER NUMBER
----------	--------------

2897

DATE MAILED: 10/19/2017

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

<i>Applicant-Initiated Interview Summary</i>	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	

All participants (applicant, applicant's representative, PTO personnel):

(1) LAWRENCE-LINH T. NGUYEN. (3)_____.

(2) Amanda Smith. (4)_____.

Date of Interview: 07/07/2017.

Type: ☒ Telephonic ☐ Video Conference
☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
If Yes, brief description: _____.

Issues Discussed ☐101 ☐112 ☒102 ☐103 ☐Others
(For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1.

Identification of prior art discussed: _.

Substance of Interview

(For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Hienrich et al., failed to teach depositing a refractory metal layer over the first intermix region of claim 1. Further search will required.

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☒ Attachment

/LAWRENCE-LINH T NGUYEN/
Examiner, Art Unit 2897

Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

It is the responsibility of the applicant or the attorney or agent to make the substance of an interview of record in the application file, unless the examiner indicates he or she will do so. It is the examiner's responsibility to see that such a record is made and to correct material inaccuracies which bear directly on the question of patentability.

Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

The Interview Summary Form shall be given an appropriate Paper No., placed in the right hand portion of the file, and listed on the "Contents" section of the file wrapper. In a personal interview, a duplicate of the Form is given to the applicant (or attorney or agent) at the conclusion of the interview. In the case of a telephone or video-conference interview, the copy is mailed to the applicant's correspondence address either with or prior to the next official communication. If additional correspondence from the examiner is not likely before an allowance or if other circumstances dictate, the Form should be mailed promptly after the interview rather than with the next official communication.

The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

A complete and proper recordation of the substance of any interview should include at least the following applicable items:

- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,
- 2) an identification of the claims discussed,
- 3) an identification of the specific prior art discussed,
- 4) an identification of the principal proposed amendments of a substantive nature discussed, unless these are already described on the Interview Summary Form completed by the Examiner,
- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
- 6) a general indication of any other pertinent matters discussed, and
- 7) if appropriate, the general results or outcome of the interview unless already described in the Interview Summary Form completed by the examiner.

Examiners are expected to carefully review the applicant's record of the substance of an interview. If the record is not complete and accurate, the examiner will give the applicant an extendable one month time period to correct the record.

Examiner to Check for Accuracy

If the claims are allowable for other reasons of record, the examiner should send a letter setting forth the examiner's version of the statement attributed to him or her. If the record is complete and accurate, the examiner should place the indication, "Interview Record OK" on the paper recording the substance of the interview along with the date and the examiner's initials.

Notice of Allowability	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 07/07/2017.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-14. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☒ Interview Summary (PTO-413),
Paper No./Mail Date <u>20170630</u> . | |

/LAWRENCE-LINH T NGUYEN/
Examiner, Art Unit 2897

DETAIL ACTION

Allowable Subject Matter

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. Applicant's arguments, see Response After Final, filed **07/07/2017**, with respect to independent claims **1 & 8** have been fully considered and are persuasive. The **102/103** rejection of independent claims **1 & 8** are withdrawn.
3. The following is an examiner's statement of reasons for allowance
 - a. With respect to independent claim **1**, the prior art teaches:
 - A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising: forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact..
 - But the prior arts of record do not teach or fairly suggest a limitation of: causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate
 - *Since the Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore,*

aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved. There is no reason apparent to the examiner that all the features claimed in claim 1 would be obvious over the prior art.

- Dependent claims 2-7 are respectively dependent on currently amended allowable independent claim 1. Therefore, claims 2-7 incorporate the allowable limitations of claim 1. Consequently, claims 2-7 are allowable.

b. With respect to independent claim 8, the prior art teaches:

- A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising: forming a contact opening in the dielectric layer to expose an upper - portion of each of the source and the drain; depositing a refractory metal layer over the first intermix region; and depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts..
- But the prior arts of record do not teach or fairly suggest a limitation of: depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion

of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain.

- *Since the* Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved. There is no reason apparent to the examiner that all the features claimed in claim 8 would be obvious over the prior art.

- Dependent claims **9-14** are respectively dependent on currently amended allowable independent claim **8**. Therefore, claims **9-14** incorporate the allowable limitations of claim **8**. Consequently, claims **9-14** are allowable.

c. This application is in condition for allowance except for the presence of claims **15-20** directed to an invention non-elected without traverse. Accordingly, claims **15-20** have been cancelled.

3. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance".

Conclusion

4. Any inquiry concerning this communication or earlier communications from the examiner should be directed to **LAWRENCE-LINH T. NGUYEN** whose telephone number is **(571) 272-6267** and fax number is **(571) 273-6267**. The examiner can normally be reached on M-F (Flex hours). If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, **Fernando Toledo** can be reached on **571-272-1867**. The fax phone number for the organization where this application or proceeding is assigned is **571-273-8300**.

5. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LAWRENCE-LINH T NGUYEN/

Examiner, Art Unit 2897

/Fernando L. Toledo/
Supervisory Patent Examiner, Art Unit 2897

<i>Applicant-Initiated Interview Summary</i>	Application No. 14/729,298	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	

All participants (applicant, applicant's representative, PTO personnel):

(1) LAWRENCE-LINH T. NGUYEN.

(3)_____.

(2) Amanda Smith.

(4)_____.

Date of Interview: 07/07/2017.

Type: ☒ Telephonic ☐ Video Conference
☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.

If Yes, brief description: _____.

Issues Discussed ☐101 ☐112 ☒102 ☐103 ☐Others

(For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1.

Identification of prior art discussed: _.

Substance of Interview

(For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Hiemrich et al., failed to teach depositing a refractory metal layer over the first intermix region of claim 1. Further search will required.

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☒ Attachment

/LAWRENCE-LINH T NGUYEN/
Examiner, Art Unit 2897

Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

It is the responsibility of the applicant or the attorney or agent to make the substance of an interview of record in the application file, unless the examiner indicates he or she will do so. It is the examiner's responsibility to see that such a record is made and to correct material inaccuracies which bear directly on the question of patentability.

Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

The Interview Summary Form shall be given an appropriate Paper No., placed in the right hand portion of the file, and listed on the "Contents" section of the file wrapper. In a personal interview, a duplicate of the Form is given to the applicant (or attorney or agent) at the conclusion of the interview. In the case of a telephone or video-conference interview, the copy is mailed to the applicant's correspondence address either with or prior to the next official communication. If additional correspondence from the examiner is not likely before an allowance or if other circumstances dictate, the Form should be mailed promptly after the interview rather than with the next official communication.

The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

A complete and proper recordation of the substance of any interview should include at least the following applicable items:

- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,
- 2) an identification of the claims discussed,
- 3) an identification of the specific prior art discussed,
- 4) an identification of the principal proposed amendments of a substantive nature discussed, unless these are already described on the Interview Summary Form completed by the Examiner,
- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
- 6) a general indication of any other pertinent matters discussed, and
- 7) if appropriate, the general results or outcome of the interview unless already described in the Interview Summary Form completed by the examiner.

Examiners are expected to carefully review the applicant's record of the substance of an interview. If the record is not complete and accurate, the examiner will give the applicant an extendable one month time period to correct the record.

Examiner to Check for Accuracy

If the claims are allowable for other reasons of record, the examiner should send a letter setting forth the examiner's version of the statement attributed to him or her. If the record is complete and accurate, the examiner should place the indication, "Interview Record OK" on the paper recording the substance of the interview along with the date and the examiner's initials.

Notice of References Cited	Application/Control No. 14/729,298		Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN		Art Unit 2897	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2012/0161324 A1	06-2012	Heinrich; Jens	H01L21/28518	257/757
*	B	US-2014/0027883 A1	01-2014	Hu; Yongjun Jeff	H01L21/743	257/534
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 14729298	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

CPC- SEARCHED		
Symbol	Date	Examiner
H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261.	09/14/2016	LTN

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

SEARCH NOTES		
Search Notes	Date	Examiner
Each of the CPC classes listed are searched in combination and/or delimited with text. See EAST search for details	09/14/2016	LTN

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
	Each of the CPC classes listed are searched in combination and/or delimited with text. See EAST search for details	09/14/2017	LTN

/LAWRENCE-LINH T NGUYEN/ Examiner.Art Unit 2897	
--	--

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant(s): Emre Alptekin

Conf. No.: 2837

Serial No.: 14/729,298

Art. Unit: 2897

Filed: June 3, 2015

Examiner: Nguyen, Lawrence –Lin T.

Docket No.: FIS920150066US1
(GFFT-0376-US)

Title: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME**

Examiner Nguyen, Lawrence –Lin T.
By Email: Lawrence-Linh.Nguyen@USPTO.GOV

**AGENDA FOR DISCUSSION IN TELEPHONE INTERVIEW
WITH EXAMINER**

Sir:

I. INTRODUCTORY COMMENTS

Applicant gratefully acknowledges the Examiner's courtesy in extending the opportunity to discuss the above-captioned patent application in a telephone interview scheduled for Friday, June 30, 2017 at 2:00 PM EDT. The proposed interview participants are to include Examiner Nguyen and Applicant's undersigned representative, Amanda Smith (Reg. No. 71,112).

II. AGENDA- DO NOT ENTER

A. Applicant would like to discuss the 35 U.S.C. § 102(a)(1) rejection of claim 1. Specifically, Applicant submits that, in the instant Office Action, the Examiner has failed to address Applicant's arguments submitted in the December 8, 2016 response. In the instant Office Action, the rejection of claim 1 from the Office Action dated September 26, 2016 is merely reproduced without any response to Applicant's previously submitted arguments. Applicant requests clarification on the rejections and why the Examiner did not find Applicant's arguments persuasive so that Applicant can better advance prosecution.

B. In the alternative, Applicant also proposes the following amendment to claim 1 to discuss relative to the 35 U.S.C. § 102(a)(1) rejection of claim 1.

1. (Proposed Amendment – Do Not Enter) A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;

depositing a first liner layer to conformally coat the contact opening;

~~causing~~ such that a portion of the first liner layer ~~to diffuse~~ diffuses into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;

depositing a refractory metal layer over the first intermix region; and

depositing a metal in the contact opening thereby forming the contact.

Applicant proposes this amendment to discuss with the Examiner without prejudice. Applicant does not concede to the rejections in the pending Office Action dated May 8, 2017. Applicant reserves the right to pursue broader scope in a response to the pending Office Action.

C. Applicant would like to discuss the 35 U.S.C. §103 rejection of claim 6. Specifically, Applicant would like to discuss the limitation of “performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and a bottom surface of the contact, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer” relative to Heinrich and Hu.

III. CONCLUSION

Thank you in advance for your consideration.

Respectfully submitted,

/ Amanda Smith /
Amanda Smith
Reg. No. 71,112

Date: June 29, 2017

Hoffman Warnick LLC
540 Broadway, 4th Floor
Albany, New York 12207
Phone: (518) 449-0044
Fax: (518) 449-0047

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	1	((14/729298).APP.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2017/08/09 23:24
L2	61	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/08/09 23:33
L3	377	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/08/09 23:33
L4	122	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/08/09 23:33
L5	141	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/08/09 23:33
L6	132	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/08/09 23:33
L7	11621	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES).AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/08/09 23:33
L8	199	"7229920" "8450807" "7795124"	US-PGPUB;	OR	ON	2017/08/09

		"6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502"	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			23:33
L9	26	(US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/08/09 23:33
L10	15491	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/08/09 23:33
L11	27710	L1 L2 L3 L4 L5 L6 L7 L10 L8 L9	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/08/09 23:33
S1	0	(14/729298).APP.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2016/09/12 20:02
S2	48	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S3	367	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:02
S4	104	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:03
S5	133	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB;	OR	ON	2016/09/12

			USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			20:03
S6	104	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:21
S7	9310	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES).AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 20:22
S8	55	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 21:07
S9	181	"7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:21
S10	26	(US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:22
S11	13120	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:24
S12	23052	S1 S2 S3 S4 S5 S6 S7 S11 S9 S10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2016/09/12 22:25

			IBM_TDB			
S13	1273	S12 and ((opening trench via hole) with dielectric) and ((liner or barrier) with metal) and (metal with (contact or fill))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:25
S14	3241	S12 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:25
S15	92	S12 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:26
S16	516	S13 S14	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:27
S17	28	S16 and ((opening or trench or via or hole) with dielectric) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:27
S18	11581	((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:44
S19	650	S12 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:45
S20	110	S12 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:47
S21	22	S19 S20	US-PGPUB; USPAT; USOCR; FPRS;	AND	ON	2016/09/12 22:48

			EPO; JPO; DERWENT; IBM_TDB			
S22	170152	((Gate Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/12 22:52
S23	63156	S22 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:55
S24	22129	S22 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S25	1183	S22 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 22:58
S26	2141	S22 and (("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/12 23:09
S27	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S28	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S29	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03

S30	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S31	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S32	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:03
S33	245794	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:05
S34	13131	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:12
S35	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:13

		or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.				
S36	259027	S27 S28 S29 S30 S31 S32 S33 S34 S35	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 11:15
S37	10581	S36 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:25
S38	29272	S36 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:25
S39	261	S36 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S40	356	S36 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:28
S41	6783	S36 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:29
S42	8119	S37 S38	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S43	2155	S37 S38 S41	US-PGPUB; USPAT; USOCR;	AND	ON	2016/09/13 13:29

			FPRS; EPO; JPO; DERWENT; IBM_TDB			
S44	6	S39 S40	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S45	0	S43 S44	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 13:29
S46	43	S43 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:42
S47	9	S43 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 13:43
S48	1709	S43 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 16:49
S49	78	S43 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "AlzO3" "Al.sub.zO.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3" "HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:11
S50	43406	S36 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:30

		(UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))				
S51	1853	S43 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:31
S52	1481	S48 S51	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:32
S53	62	S52 and ((Gate adj Dielectric Gate) adj3 ("HfO2" "HfO.sub.2" "ZrO2" "ZrO.sub.2" "La2O3" "La.sub.2O.sub.3" "Al2O3" "Al.sub.2O.sub.3" "TiO2" "TiO.sub.2" "SrTiO3" "SrTiO.sub.3" "LaAlO3" "LaAlO.sub.3" "Y2O3" "Y.sub.2O.sub.3"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2016/09/13 17:32

		"HfOxNy" "HfO.sub.xN.sub.y" "ZrOxNy" "ZrO.sub.xN.sub.y" "La2OxNy" "La.sub.2O.sub.xN.sub.y" "AlzOxNy" "Al.sub.zO.sub.xN.sub.y" "TiOxNy" "TiO.sub.xN.sub.y" "SrTiOxNy" "SrTiO.sub.xN.sub.y" "LaAlOxNy" "LaAlO.sub.xN.sub.y" "Y2OxNy" "Y.sub.2O.sub.xN.sub.y" "SiON" "SiNx" "SiN.sub.x" Silicate))	IBM_TDB			
S54	5	S52 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S55	29	S52 and ((Form\$3 Deposit\$3) with (Remov\$3 Extract\$3 Etch\$3) with (Refract Refractory))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:32
S57	1481	S52 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S58	1452	S52 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:39
S59	318	S52 and ((Drain and Gate and Source and Spacer) with (Trench Open Cavity Contact adj Open Contact Metal ((Metal Contact) adj Liner)) with (Dielectric Isolat\$3 Liner Mask Oxide))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/13 17:40
S60	318	S57 S58 S59	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/13 17:40
S61	16	("20010035558" "20050124128" "20080171411" "20080272357" "20080303068" "20100075474" "6087209" "6093243" "6440828" "6812082" "7259110" "7736964" "7755114" "7785947").PN. OR ("8304819").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:07
S62	21	("20040056281" "20040121590" "20050020093" "20050026380" "20060141712" "20060166494"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:15

		"20070042580" "20070141798" "20090134497" "20110084320" "20120202326" "20120248550" "20140015104" "6323519" "6730570" "7087515" "7220668" "7348283" "7679192" "8304819" "8492217").PN. OR ("9252019").URPN.				
S63	2	"20130049219"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:43
S64	21	("20040056281" "20040121590" "20050020093" "20050026380" "20060141712" "20060166494" "20070042580" "20070141798" "20090134497" "20110084320" "20120202326" "20120248550" "20140015104" "6323519" "6730570" "7087515" "7220668" "7348283" "7679192" "8304819" "8492217").PN. OR ("9252019").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:44
S65	13	("20040067658" "20040150075" "20050110153" "20060003577" "6878620" "7087515" "7199044" "7253097" "7253524" "7309658" "7466027").PN. OR ("7679192").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:49
S66	4	"20130049219" "20070152334"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/14 14:51
S67	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S68	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S69	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S70	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S71	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2016/09/14 15:09

			DERWENT; IBM_TDB			
S72	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S73	245794	("GLOBALFOUNDRI ES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S74	13131	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S75	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S76	259027	S67 S68 S69 S70 S71 S72 S73 S74 S75	US-PGPUB; USPAT; USOCR;	OR	ON	2016/09/14 15:09

			FPRS; EPO; JPO; DERWENT; IBM_TDB			
S77	10581	S76 and ((Drain Gate Source Spacer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 15:09
S78	29272	S76 and ((Opening Open\$3 Trench Via Cavity Hole) with (Dielectric Insulat\$3 Isolat\$3 IDL Interlayer adj Dielectric))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S79	6783	S76 and ((Fusing Fused Fussion Diffuse Diffus\$3) with (Liner Dop\$3 Intermix\$3 Refract\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S80	2155	S77 S78 S79	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 15:09
S81	1709	S80 and ((silicon "Si" germanium "Ge" silicon adj germanium "SiGe" "Si.sub" adj "Ge.sub" silicon adj carbide "SiC" "III-V") near3 (Substrate submount wafer))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S82	1853	S80 and ((chemical adj vapor adj deposition "chemical vapor deposition" (CVD) "low-pressure CVD" (LPCVD) "plasma-enhanced CVD" (PECVD) "semi-atmosphere CVD" (SACVD) "high density plasma CVD" "high density plasma" (HDPCVD) "rapid thermal CVD" (RTCVD) "ultra-high vacuum CVD" (UHVCVD) "limited reaction processing CVD" (LRPCVD) "metalorganic CVD" (MOCVD) ((sputtering ion adj beam electron adj beam laser adj assisted physical adj vapor atomic adj layer) adj (Deposit deposition)) "electron beam deposition" "sputtering deposition" "ion beam deposition" "laser assisted deposition" "thermal oxidation" "thermal nitridation spin-on" "physical vapor deposition" (PVD) "atomic layer deposition" (ALD) "chemical oxidation" "molecular beam epitaxy" (MBE) plat\$3 evaporat\$3) ((low-pressure low adj pressure plasma-enhanced plasma adj	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09

		enhanced semi-atmosphere semi adj atmosphere high adj density adj plasma rapid adj thermal ultra-high adj vacuum ultra adj high adj vacuum limited adj reaction adj processing metalorganic) adj (CVD)))				
S83	1481	S81 S82	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 15:09
S84	34	S83 and ((("W" Tungsten "Ti" Titanium "Ta" Tantalum "Mo" Molybdenum "TiN" Titanium adj Nitride "TaN" Tantalum adj Nitride Titanium-Tungsten Titanium adj Tungsten) with ("W" Tungsten "Co" Colbalt "Cu" Copper and "Al" Aluminum) with ("NiPt" Nickel-Platinum Nickel adj Platinum "Ti/TiN" bilayer Liner/Barrier Liner adj2 Barrier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 15:09
S85	8119	S77 S78	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 17:08
S86	2155	S77 S78 S79	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2016/09/14 17:08
S87	407	S76 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) with (Remov\$3 Extract\$3 Etch\$3) with (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 17:35
S88	45	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) with (Remov\$3 Extract\$3 Etch\$3) with (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 17:37
S89	0	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 20:05
S90	118	S86 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Source Drain)	US-PGPUB; USPAT; USOCR; FPRS;	OR	ON	2016/09/14 20:06

		)	EPO; JPO; DERWENT; IBM_TDB			
S91	33	S90 and ((Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 20:06
S92	1480	S80 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:52
S93	22	(US-20150221740-\$ or US- 20130240990-\$ or US-20100078771-\$ or US-20090023261-\$ or US- 20120161324-\$ or US-20130119483-\$ or US-20140319620-\$ or US- 20140246698-\$ or US-20120104498-\$ or US-20080040697-\$ or US- 20060163631-\$ or US-20070122956-\$ or US-20130049219-\$ or US- 20070152334-\$).did. or (US-8304819-\$ or US-9224675-\$ or US-8742510-\$ or US-9093425-\$ or US-9252019-\$ or US- 6440828-\$ or US-7679192-\$).did. or (US-20110084320-\$).did.	US-PGPUB; USPAT; DERWENT	OR	ON	2016/09/14 21:52
S94	16	S93 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:52
S95	4	S93 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/14 21:57
S96	22455	H01L21/7684 H01L21/76841 H01L21/76843 H01L21/76846 H01L21/7685 H01L21/76879 H01L21/76883 H01L23/481 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 10:32
S97	69	"20150187896" "5565708"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 10:33
S98	639	S96 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical	US-PGPUB; USPAT;	OR	ON	2016/09/15 12:44

		Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S99	21	S98 and ((open gap void cavity opening or trench or via or hole) with (insulat\$3 isolat\$3 IDL dielectric Interlayer adj Dielectric)) and ((two near2 metal near3 liner) or (first near3 metal near3 liner) or (second near3 metal near3 liner))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:44
S100	1	"14729298"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S101	49	((Nicolas) near2 (Breil)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S102	368	((Christian) near2 (Lavoie)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S103	105	((Ahmet) near2 (Ozcan)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S104	134	((Kathryn) near2 (Schonenberg)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S105	105	((Emre) near2 (Alptekin)).INV.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S106	245815	("GLOBALFOUNDRIES INC." GLOBALFOUNDRIES "IBM" International adj Business adj Machines "International Business Machines").AS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45

S107	13151	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S108	276	("8278200").PN. ("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN. "7229920" "8450807" "7795124" "6624489" "7785972" "20110297911" "20120112292" "20120326317" "20070249149" "20110147798" "20110221003" "20130240990" "20130009217" "20090289302" "20140027826" "20110147870" "20120129302" "6756651" "7157300" "7262117" "7566875" "7616904" "8765502" (US-20130240990-\$ or US- 20120187460-\$ or US-20120252198-\$ or US-20110221003-\$ or US- 20130009217-\$ or US-20090289302-\$ or US-20140145319-\$ or US- 20140138837-\$ or US-20090302480-\$ or US-20100078771-\$ or US- 20050085064-\$ or US-20060246656-\$ or US-20120132966-\$) .did. ("20070059878" "7253071" "8101518" "20070181954" "20070093047" "20020111021" "6346465" "20130119483" "20070134898" "6468901" "20080121950" "20100224938" "20120074501" "20070202692" "8338239").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S109	259067	S100 S101 S102 S103 S104 S105 S106 S107 S108	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:45
S110	19	S109 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 12:57
S111	20386	S109 and (chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP) "laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	OR	ON	2016/09/15 13:52

			IBM_TDB			
S112	4016	S109 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 13:53
S113	0	S98 and ((intrinsic purity impurities Unreacted "non-diffused" "not diffused" (("Not" "Un" "Non") adj (Diffuse Diffus\$3 React\$3))) adj5 (Remov\$3 Extract\$3 Etch\$3) adj5 (Source Drain))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 13:54
S114	0	S112 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:00
S115	84	S109 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:01
S116	0	S98 and ((First Second Top Bottom Upper Lower High\$3 Low\$3 Left Right Adjacent) adj (refractory adj (material metal)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:01
S117	34	("20020019107" "20020102810" "20040206620" "20040259308" "20050087879" "20050263890" "20070145525" "20080073708" "20080138948" "20080188055" "20080272421" "20100001325" "20100025853" "20100187655" "20100213571" "20110057292" "20130011990" "5981331" "6219125" "6511879" "6593616" "6746916" "6753618" "6881642" "6906420" "7071054" "7268072" "7304367" "7329939" "7368343" "7545009" "7772678" "8105891").PN. OR ("8835274").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/15 14:04
S118	2	"20140027883"	US-PGPUB; USPAT; USOCR	OR	ON	2016/09/15 14:04
S119	0	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) and ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:11

S120	0	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) with ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:12
S121	36	S115 and ((chemical adj mechanical adj polish\$3 chemical adj mechanical Polish\$3 "chemical mechanical polishing" (CMP)) ("laser assisted deposition" Laser adj Anneal\$3 "Laser Anneal" Laser))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2016/09/15 14:12
S122	1	(14/729298).APP.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2017/04/17 13:15
S123	53	("20070249149" "20110147798" "20120112292" "20120181697" "8304819" "7229920" "20100320547" "5726083" "6624489" "20110297911" "7795124" "8450807" "20100078771" "6440828" "20120326317" "7785972" "20090023261" "20130240990").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/04/17 13:33
S124	1	(14/729298).APP.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2017/06/30 13:36
S125	1	(14/729298).APP.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2017/06/30 13:39
S126	2	("20120161324").PN.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2017/06/30 14:00

Issue Classification 	Application/Control No. 14729298	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897	

CPC						
Symbol					Type	Version
H01L		23		53266	F	2013-01-01
H01L		21		76843	I	2013-01-01
H01L		21		76846	I	2013-01-01
H01L		21		76879	I	2013-01-01
H01L		21		7684	I	2013-01-01
H01L		21		76883	I	2013-01-01
H01L		23		485	I	2013-01-01
H01L		23		53223	A	2013-01-01
H01L		23		53238	A	2013-01-01

CPC Combination Sets									
Symbol					Type	Set	Ranking	Version	

/LAWRENCE-LINH T NGUYEN/ Examiner.Art Unit 2897	08/10/2017	Total Claims Allowed: 14	
(Assistant Examiner)	(Date)		
/FERNANDO L TOLEDO/ Supervisory Patent Examiner.Art Unit 2897	08/21/2017	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	9

Issue Classification 	Application/Control No. 14729298	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1														
	2														
	3														
	4														
	5														
	6														
	7														
	8														
	9														
	10														
	11														
	12														
	13														
	14														

/LAWRENCE-LINH T NGUYEN/ Examiner.Art Unit 2897		08/10/2017		Total Claims Allowed:	
(Assistant Examiner)		(Date)		14	
/FERNANDO L TOLEDO/ Supervisory Patent Examiner.Art Unit 2897		08/21/2017		O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)		(Date)		1	9

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor: Emre Alptekin	Conf. No.: 2837
Serial No.: 14/729,298	Art. Unit: 2897
Filed: June 3, 2015	Examiner: Nguyen, Lawrence-Lin T.
	Docket No.: FIS920150066US1 (GFFT-0376-US)
Title: CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME	

STATEMENT OF SUBSTANCE OF INTERVIEW

This paper is being filed in response to the Notice of Allowance and Applicant-Initiated Interview Summary mailed October 19, 2017. Applicant would like to clarify that the telephone interview between Examiner Nguyen and Applicant's representative, Amanda J. Smith (Reg. No. 71,112), prior to the Notice of Allowance was conducted on June 30, 2017, *not* July 7, 2017 as indicated by the Office in the Applicant-Initiated Interview Summary mailed October 19, 2017. It is further noted that an After-Final Response, including an Interview Summary on page 6, was previously filed on July 7, 2017, and includes remarks on the interview of June 30, 2017.

Respectfully submitted,

/ Justine M. Wilbur /
Justine M. Wilbur
Reg. No. 59,678

Date: December 5, 2017

Hoffman Warnick LLC
540 Broadway, 4th Floor
Albany, New York 12207
Phone: (518) 449-0044
Fax: (518) 449-0047

14/729,298
FIS920150066US1

Electronic Patent Application Fee Transmittal

Application Number:	14729298			
Filing Date:	03-Jun-2015			
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME			
First Named Inventor/Applicant Name:	Emre Alptekin			
Filer:	Justine M. Wilbur/Deanna Cole			
Attorney Docket Number:	FIS920150066US1			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
UTILITY APPL ISSUE FEE	1501	1	960	960
PUBL. FEE- EARLY, VOLUNTARY, OR NORMAL	1504	1	0	0
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				960

Electronic Acknowledgement Receipt

EFS ID:	31125803
Application Number:	14729298
International Application Number:	
Confirmation Number:	2837
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Justine M. Wilbur
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US1
Receipt Date:	05-DEC-2017
Filing Date:	03-JUN-2015
Time Stamp:	12:08:29
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$960
RAM confirmation Number	120517INTEFSW12092200
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	FIS920150066US1_IssueFeeTransmittal_12-05-17.pdf	110179	no	1
			55a914e543fef037bc7eb8d17bbaf196bf79a30b		
Warnings:					
Information:					
2	Maintenance Fee Address Change	FIS920150066US1_FeeAddressIndicationForm_12-05-17.pdf	312573	no	2
			108d98f98b9c6cff377d5daa886c1efe557840fa		
Warnings:					
Information:					
3	Supplemental Response or Supplemental Amendment	FIS920150066US1_StatementOfSubstanceOfInterview_12-05-17.pdf	17893	no	1
			39c310be5e5aae982330ae88f3f468fdeb901335		
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	32560	no	2
			2e898d629ad9472bd12cc655e37faacc0f3128e0		
Warnings:					
Information:					
Total Files Size (in bytes):			473205		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

133964 7590 10/19/2017
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/729,298	06/03/2015	Emre Alptekin	FIS920150066US1	2837

TITLE OF INVENTION: CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	01/19/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
NGUYEN, LAWRENCE-LIN T	2897	257-757000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☒ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Yuanmin Cai

2 Hoffman Warnick LLC

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

GLOBALFOUNDRIES INC.

Grand Cayman, KY

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

☒ Issue Fee

☒ Publication Fee (No small entity discount permitted)

☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

☐ A check is enclosed.

☒ Payment by credit card. Form PTO-2038 is attached.

☒ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number 500999 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Justine M. Wilbur/

Date Decmeber 5, 2017

Typed or printed name Justine M. Wilbur

Registration No. 59,678

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

"FEE ADDRESS" INDICATION FORM

Address to:
Mail Stop M Correspondence
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Fax to:
571-273-6500

- OR -

INSTRUCTIONS: The issue fee must have been paid for application(s) listed on this form. In addition, only an address represented by a Customer Number can be established as the fee address for maintenance fee purposes (hereafter, fee address). A fee address should be established when correspondence related to maintenance fees should be mailed to a different address than the correspondence address for the application. **When to check the first box below:** If you have a Customer Number to represent the fee address. **When to check the second box below:** If you have no Customer Number representing the desired fee address, in which case a completed Request for Customer Number (PTO/SB/125) must be attached to this form. For more information on Customer Numbers, see the Manual of Patent Examining Procedure (MPEP) § 403.

For the following listed application(s), please recognize as the "Fee Address" under the provisions of 37 CFR 1.363 the address associated with:



Customer Number:

00197**OR**

The attached Request for Customer Number (PTO/SB/125) form.

PATENT NUMBER (if known)	APPLICATION NUMBER
	14/729,298

Completed by (check one):



Applicant/Inventor

/Justine M. Wilbur/

Signature



Attorney or Agent of record 59,678
 (Reg. No.)

Justine M. Wilbur

Typed or printed name



Assignee of record of the entire interest. See 37 CFR 3.71.
 Statement under 37 CFR 3.73(b) is enclosed.
 (Form PTO/SB/96)

518-449-0044

Requester's telephone number



Assignee recorded at Reel _____ Frame _____

December 5, 2017

Date

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below*.

* Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.363. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 5 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND COMPLETE D FORMS TO THIS ADDRESS. **SEND TO: Mail Stop M Correspondence, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/729,298	01/09/2018	9865546	FIS920150066US1	2837

133964 7590 12/20/2017
HOFFMAN WARNICK LLC
540 Broadway
4th Floor
ALBANY, NY 12207

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 30 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Emre Alptekin, Wappingers Falls, NY;
GLOBALFOUNDRIES INC., Grand Cayman, CAYMAN ISLANDS;
Nicolas L. Breil, San Jose, CA;
Christian Lavoie, Pleasantville, NY;
Ahmet S. Ozcan, Chappaqua, NY;
Kathryn T. Schonenberg, Wappingers Falls, NY;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.